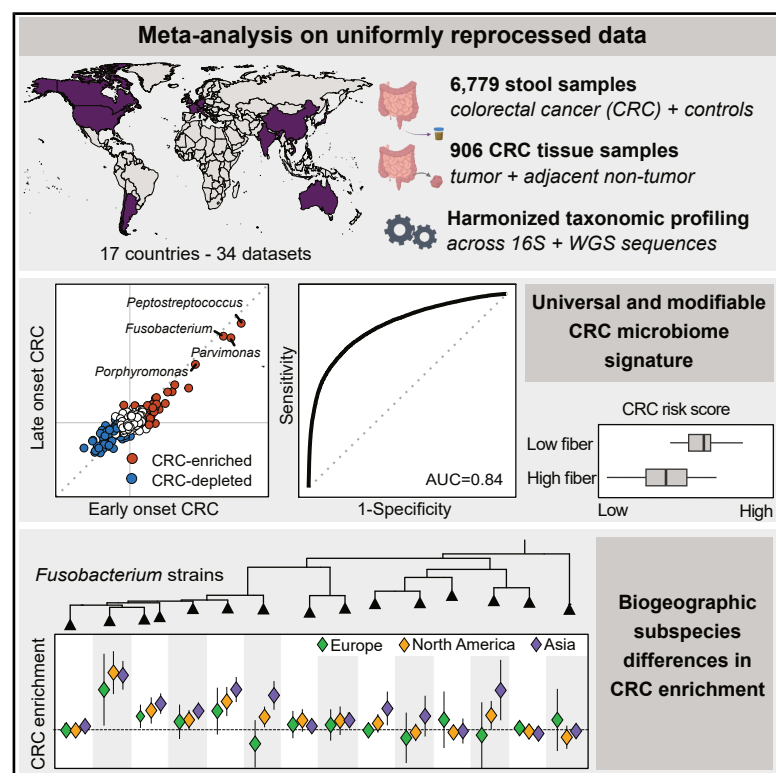


Cell Host & Microbe

Meta-analysis reveals microbiome signatures for colorectal cancer that are universal across age groups and sequencing methods

Graphical abstract



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In brief

Pekel, Karcher, et al. present the largest gut microbiome meta-analysis of colorectal cancer to date, integrating 6,779 amplicon and shotgun sequencing profiles. They establish a robust microbiome signature across populations, sequencing approaches, and age of onset; associate it with dietary fiber intake; and reveal the biogeography of cancer-enriched *Fusobacterium* subspecies.

Highlights

- Gut microbiome CRC signature is universal across age groups and sequencing approaches
- Tumor-enriched microbes are concordant with fecal CRC signatures
- Increasing dietary fiber intake reduces the microbiome CRC signature score
- *Fusobacterium* subspecies show biogeographic variation in CRC enrichment



Article

Meta-analysis reveals microbiome signatures for colorectal cancer that are universal across age groups and sequencing methods

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SUMMARY

Numerous studies have linked gut microbiome alterations to colorectal cancer (CRC), but limited sample sizes and study heterogeneity have hampered cross-study comparisons and subgroup analyses. Here, we present a comprehensive single-disease gut microbiome meta-analysis based on consistently re-computed and re-analyzed shotgun and amplicon sequencing profiles ($n = 6,779$ samples, 27 studies). Association and machine-learning analyses delineate CRC microbiome signatures, which are robustly generalizable across studies and sequencing approaches and nearly identical between early- and late-onset cases. Meta-analysis of the tumor-resident microbiome reveals characteristic tumor-enriched microbes in concordance with fecal signatures that are clearly detectable in early-stage tumors, although their detection in feces becomes moderately higher in late-stage and distal tumors, possibly due to dilution effects in stool. The unified fecal CRC signature inversely associates with dietary fiber intake and is modifiable by dietary interventions. Finally, genome-resolved functional analysis reveals variation in virulence factor carriage and geographic enrichment across *Fusobacterium* subspecies.

INTRODUCTION

The human gut microbiome is increasingly recognized as an important factor in the initiation and progression of colorectal cancer (CRC).^{1–4} Microbiome surveys of stool and colon tissue have played a major role in defining polymicrobial signatures of CRC and guiding our understanding of tumor-associated communities.^{5–11} These surveys found numerous bacterial species to be enriched in CRC, including *Fusobacterium nucleatum*, *Parvimonas micra*, *Peptostreptococcus stomatis*, *Porphyromonas asaccharolytica*, and *Gemella morbillorum*, while beneficial, short-chain fatty acid-producing taxa are often depleted.^{5,8,9,12,13} Subsequent experimental work using model systems has shown that candidate bacteria such as *F. nucleatum* can initiate tumorigenesis by interactions with host epithelial cells mediated by adhesins and toxin delivery.^{14–18} Additionally, indirect bacterial impact on CRC progression via immune modulation and (local) inflammation is also well documented.^{19,20}

Despite the progress made in delineating CRC-associated microbial signatures from metagenomic data,^{5–11,21} important questions remain concerning the robustness and generalization of these signatures given the technical and biological heterogeneity across studies^{8,9,22}: what is the impact of differences in sample collection, DNA extraction, and sequencing approaches?²³ Could geographic and demographic differences between studies confound CRC signatures? How universal are CRC signatures across cancer subtypes and stages? Are gut microbiome changes in early-onset CRC (EO-CRC) different from those in the more prevalent late-onset CRC (LO-CRC)? EO-CRC is of particular interest due to its rising incidence and challenge for early detection. Recent studies comparing EO-CRC to LO-CRC in terms of gut microbiome alterations suggested the existence of a specific EO-CRC gut microbial signature.^{22,24} However, these findings remain geographically restricted and were challenged by other studies,²⁵ calling for further investigation in diverse populations.



Examining CRC subgroups to address these questions requires either specific study designs or large-scale meta-analyses. A lack of bioinformatics standards in the microbiome field necessitates meta-analyses that uniformly re-profile raw microbiome data to reduce technical heterogeneity and that employ statistical frameworks that account for remaining sources of variation.^{26–28} While some progress has been made toward joint analysis of 16S rRNA gene amplicon (abbreviated 16S in the following) and whole-genome shotgun (WGS) sequencing data, it remains challenging to combine data from both sequencing approaches for the delineation of microbiome disease signatures.²⁹

Here, we present a comprehensive CRC meta-analysis of gut microbiomes, in which we uniformly reprofiled 1,948 WGS and 4,831 16S fecal microbiome samples as well as 906 tumor tissue samples using a consistent framework for statistical and machine-learning (ML) analysis. We found the gut microbiome CRC signature resulting from this meta-analysis to be remarkably robust to study heterogeneity and strongly reflecting tumor tissue-enriched microbes. We unified this signature into a statistical classifier that accurately discriminated CRC- versus control-derived fecal microbiome profiles and explored how detection accuracy varied across age of onset, tumor location, stage, and precursor lesions. We further investigated how CRC classification scores are associated with dietary patterns and can be modulated by dietary interventions. Finally, we conducted a functional meta-analysis of microbial virulence factors (VFs), as well as a genomic, strain-resolved analysis exploring geographic differences in CRC association between *Fusobacterium* subspecies.

RESULTS

A gut microbial CRC signature that is consistent across studies and sequencing approaches

To globally investigate gut microbial signatures of CRC, we acquired public data from 6,779 case-control stool samples from 27 published studies across 15 countries, identified through a systematic PubMed search (Tables S1 and S2; see STAR Methods). This included both WGS and 16S data generated at the time of diagnosis and before therapy. For these, we re-computed taxonomic profiles in a framework facilitating integrative analysis of both data types (STAR Methods). This resulted, to our knowledge, in the largest microbiome dataset for a single disease analyzed to date.

We first assessed overall variation in microbiome composition across studies and sequencing approaches using principal coordinates analysis (PCoA) of genus-level relative abundances (Figures 1A and S1). Study-specific differences explained a large portion of community structure (Bray-Curtis, PERMANOVA $R^2 = 0.18$, $p < 0.001$; Figure 1A), in line with previous findings.^{8,9} Among 16S datasets specifically, the hypervariable region targeted by PCR induced noticeable segregation (Bray-Curtis, PERMANOVA $R^2 = 0.096$, $p < 0.001$; Figure S1). In comparison, compositional differences between CRC cases and controls (in the combined dataset) were more subtle but significant (PERMANOVA $R^2 = 0.003$, $p < 0.001$; Figure S1). Altogether, this analysis motivated the use of downstream statistical methodology which accounted for study heterogeneity.

To assess if the structural community differences observed between sequencing approaches also affected associations of individual gut microbial taxa with CRC, we conducted separate meta-analyses of 347 prevalent bacterial genera for 16S ($n = 4,831$) and WGS ($n = 1,948$) profiles using a linear mixed effect model (LMM) framework accounting for study heterogeneity (Table S3; STAR Methods). For both sequencing approaches, we observed an enrichment of previously described CRC-associated bacteria—including *Peptostreptococcus*,^{30,31} *Parvimonas*,^{7,32,33} *Fusobacterium*,^{7–9} and *Porphyromonas*^{34,35}—and depletion of health-associated genera involved in short-chain fatty acid production, such as *Lachnospira*³⁶ (Figures 1B, 1C, and S1). Despite the observed study heterogeneity, consistency between CRC-associated genera was remarkably high between 16S and WGS datasets (Figure 1D; Pearson's R^2 of effect size estimates = 0.65, Jaccard index = 0.41).

We next investigated whether ML classifiers could leverage microbiome differences to discriminate between CRC cases and controls and if these models exploited sequencing method-specific taxonomic features. Specifically, we separately cross-validated random forest classifiers on normalized genus-level relative abundance profiles from either 16S or WGS datasets (see STAR Methods). For both data types, classifiers achieved high within-assay prediction accuracy (area under the curve [AUC] = 0.80 for 16S; 0.81 for WGS). Interestingly, when applying these models across data types (i.e., testing WGS data with a 16S-based model that had not seen any WGS data during training and vice versa), they maintained high prediction accuracy (AUC ≥ 0.78 ; Figure 1E). Similar to our comparison of univariate genus associations, these results indicate that ML models capture a universal CRC signature that is only minimally affected by differences in microbiome profiles generated by different approaches (e.g., 16S primer bias³⁷ for different amplicons).

Microbial signatures of EO-CRC and LO-CRC are highly similar

Based on the observation that CRC signatures were robust to study heterogeneity, we proceeded to study relevant subgroups of CRC on a pooled dataset (see Figure 1). Among these subgroups, EO-CRC is of primary interest due to its globally rising incidence.^{38,39} We investigated EO-CRC-associated microbiome changes at a far larger scale than previous studies reporting distinct signatures in EO-CRC and LO-CRC based on patient populations from (eastern) China.^{22,24,25} For this analysis, we stratified CRC cases into EO-CRC (age at diagnosis <50 years) and LO-CRC (age ≥ 50 years) groups across a total of 22 datasets with available annotations for age of onset (total $n = 5,663$; Table S1). We separately conducted meta-analyses of genus-level associations with LO-CRC and EO-CRC, based on the same LMM approach as described above. Interestingly, this revealed highly consistent microbial enrichment patterns for EO-CRC and LO-CRC, with similar directionality and effect sizes of microbial differences across studies (Figures 2A and S2; Table S4). Notably, the genera most strongly enriched in CRC, *Peptostreptococcus*, *Fusobacterium*, *Parvimonas*, and *Porphyromonas*, were particularly consistent irrespective of age of onset (Figure 2B), and effect sizes between EO-CRC

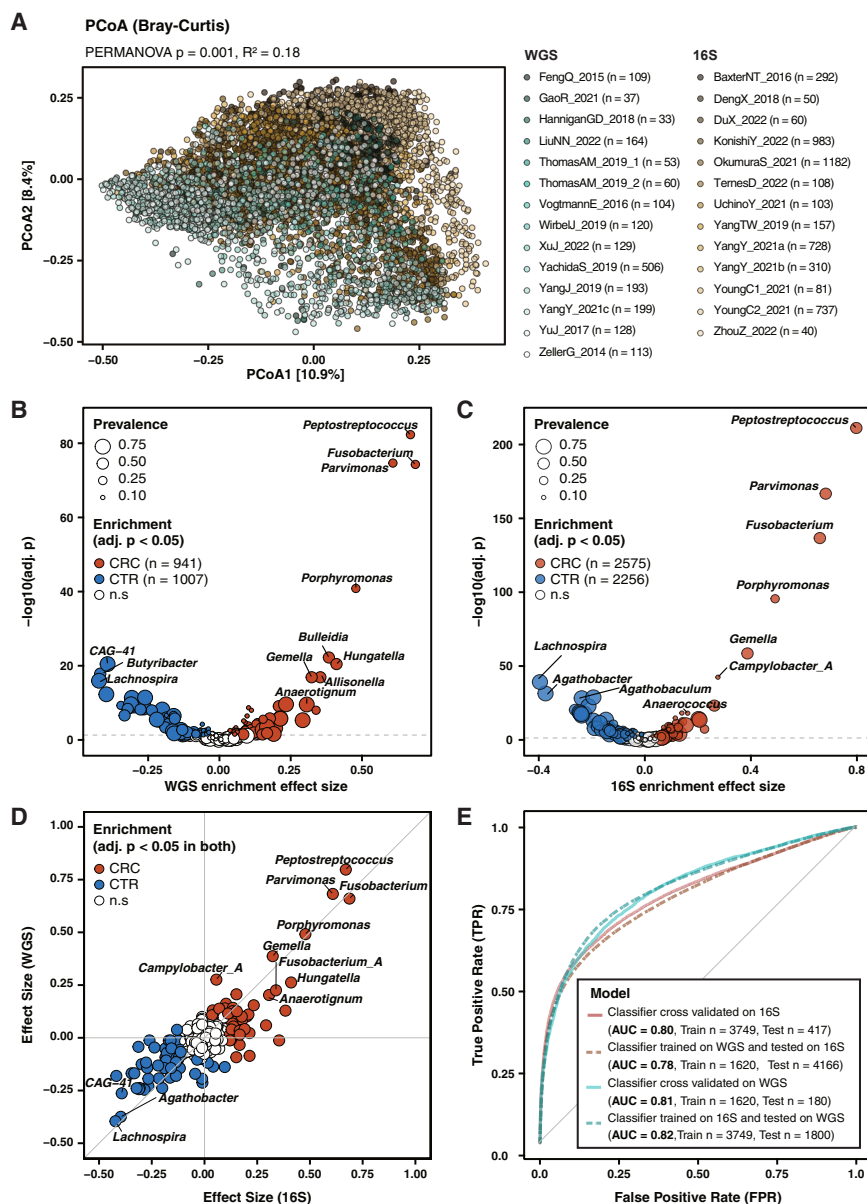


Figure 1. Gut microbial signatures of CRC are highly consistent across sequencing approaches

(A) PCoA of CRC cases and control samples used in this study ($n = 6,779$).

(B) Meta-analysis for differential abundance among 347 genera in WGS profiles of CRC cases and controls (CTR) ($n = 1,948$ from 14 datasets) was performed using LMMs to account for study heterogeneity (see Table S3 and STAR Methods).

(C) Similar meta-analysis as in (B), performed on 16S profiles ($n = 4,831$ from 13 datasets).

(D) Comparison of genus-level CRC enrichments between 16S and WGS profiles shown in (B) and (C). Colored taxa were observed as significant in both analyses ($FDR \leq 0.05$).

(E) Receiver operating characteristic (ROC) curves comparing the performance of classifiers cross-validated on 16S and WGS data as well as classifiers trained on one sequencing data type and evaluated on the other with the AUC (see legend and STAR Methods). Train and test sample sizes were averaged per cross-validation (CV) fold.

omonas; Figure 2C). Differences between EO-CRC and LO-CRC in terms of gut microbial genera were not sensitive to the age cutoff used to delineate EO-CRC and LO-CRC (Figure S2; Table S4). Overall, these analyses underscore that CRC-associated microbiome alterations are restricted to few taxa where the disease association clearly predominates over age-related differences.

To further corroborate the similarity of microbiome signatures between EO-CRC and LO-CRC, we also tested the transferability of ML models across age groups. To this end, we independently trained classifiers on the EO-CRC and LO-CRC datasets and evaluated the classifiers on the other (held-out) age group. Generally, ML classifiers tend to be sensitive

and LO-CRC were overall highly correlated (Pearson's $R^2 = 0.82$, Jaccard index = 0.48).

Moreover, when directly testing for differences between EO-CRC and LO-CRC using LMMs, we identified several differentially abundant genera, albeit at very small effect sizes (Figure S2). Consistent with this, we observed substantial overlap between EO- and LO-CRC samples in beta diversity analysis, with age of onset explaining only a small proportion of overall variance (Figure S2). Shannon diversity did not significantly change between EO- and LO-CRC nor between EO- and LO-CTR (control) after accounting for study differences. When directly contrasting the variance in genus abundances explained by either CRC status or age of onset, we found that CRC status accounted for substantially more microbiome variation than age (between 8%–16% and <1%, respectively, for the key genera *Peptostreptococcus*, *Fusobacterium*, *Parvimonas*, and *Porphy-*

even to subtle differences in feature distributions; hence, if enrichment patterns differed between EO-CRC and LO-CRC (relative to the respective control populations), a classifier trained on LO-CRC would be expected to show poor predictivity on EO-CRC. However, here we observed similarly strong predictive accuracy for classifiers cross-validated on only EO-CRC (and age-appropriate controls, AUC of 0.79) as compared with exclusively LO-CRC-trained classifiers evaluated on EO-CRC (AUC of 0.82; Figures 2D and S2). LO-CRC classifier performance did not vary substantially when using different age cut-offs (Figure S2; Table S4). To identify the microbial taxa most strongly contributing to CRC classification in the respective EO-CRC and LO-CRC ML models, we quantified feature contributions using Shapley additive explanation (SHAP) values (see STAR Methods). This analysis revealed similar taxa driving predictions in LO-CRC and EO-CRC ML

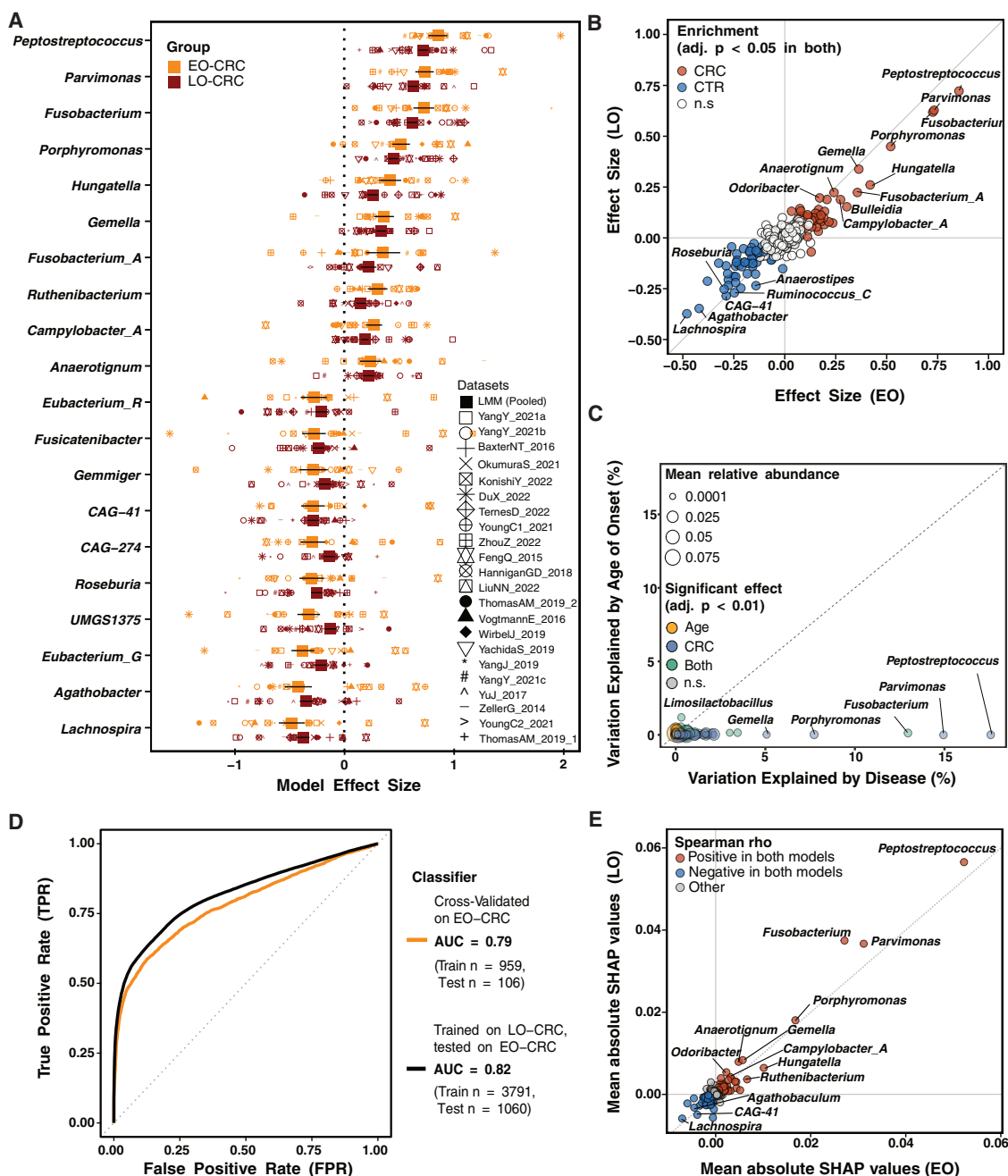


Figure 2. Gut microbial signatures of EO-CRC and LO-CRC are highly similar

(A) Forest plot showing the differentially abundant genera between CRC and CTR samples along with corresponding 95% confidence intervals and study-specific estimates calculated for both EO-CRC and LO-CRC using LMMs (see Table S4). Only the 10 most strongly CRC- and CTR-associated genera are shown. (B) Scatterplot comparing overall genus-level CRC enrichment estimates between EO- and LO-CRC. (C) Variance in bacterial abundances explained by disease status (x axis) and age at diagnosis (y axis), obtained from partial R^2 estimates (STAR Methods). (D) ROC curves of CRC classifiers cross-validated on EO-CRC samples (orange) or trained on all LO-CRC samples and tested on all EO-CRC samples (black). Train and test sample sizes are averages per cross-validation (CV) fold. (E) Scatterplots comparing genus-level mean absolute SHAP values of EO-CRC and LO-CRC random forest models (STAR Methods).

models, irrespective of age group, with genera such as *Fusobacterium*, *Peptostreptococcus*, and *Parvimonas* ranking among the most influential features (Figure 2E), in close agreement with univariate genus-level associations (see Figures 2A and 2B).

A unified microbiome classifier accurately detects CRC across global populations

Having demonstrated that ML classifiers generalize well across sequencing approaches and age of onset, we next sought to construct a unified CRC microbiome signature. This unified

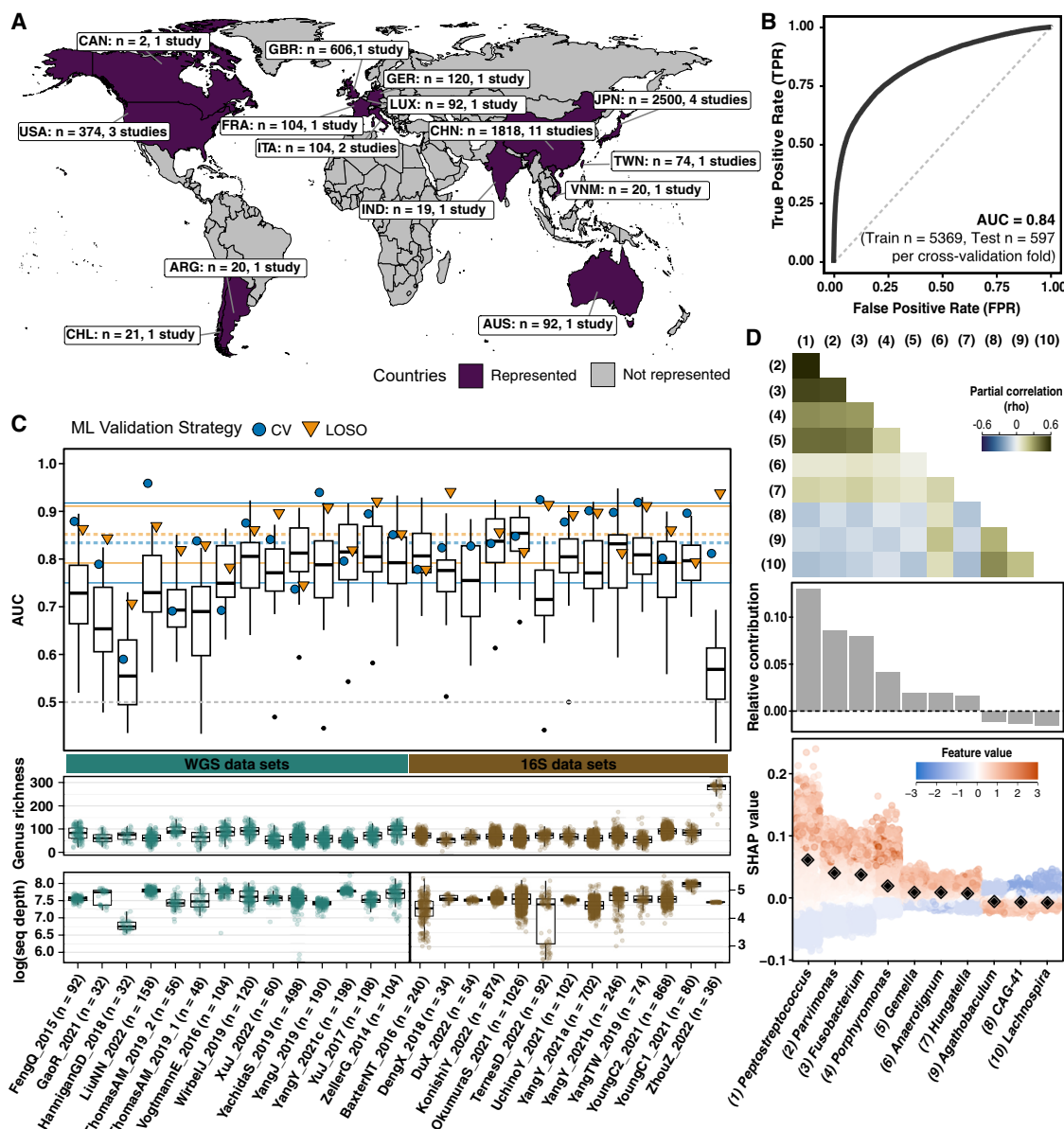


Figure 3. A unified classifier recognizes CRC across fecal microbiome profiles from diverse populations

(A) Geographic distribution and sample sizes (n) of studies included in the unified CRC ML model.
(B) ROC curve showing the performance of the unified CRC model in distinguishing CRC (n = 2,983) from CTR (n = 2,983).
(C) Performance of cross-validation (CV) and LOSO models across individual 16S and WGS datasets (see Table S5). Each boxplot summarizes the distribution of AUC values from study-to-study transfer validation, where models trained on one study were tested on others individually. Additionally, blue circles indicate AUC values from within-study CV, and orange triangles represent AUC values from LOSO validation, in which models were trained on all datasets except the held-out study shown on the x axis. Horizontal lines denote the overall mean (dashed) and standard deviation (solid line) of AUCs for both LOSO (orange) and CV (blue). Underneath, genus richness and sequencing depths are summarized for each study. Boxplots denote first and third quartiles with a median line in the center and whiskers extending to 1.5 times the interquartile range.
(D) Dot plot of SHAP values computed on the unified CRC model. Individual dots (bottom) correspond to a feature's SHAP value in a sample, with the mean indicated by solid diamonds. The accompanying bars show the mean absolute SHAP values, scaled to represent each feature's relative contribution to the model. The triangular heatmap on top shows pairwise partial Spearman correlations between these marker genera, estimated from centered log-ratio transformed relative abundances and a linear model adjusted for disease status and study of origin. Only the top 10 features (by relative contributions to the model) are shown.

random forest classifier was cross-validated on genus-level relative abundances derived from both 16S and WGS data across 27 studies and 15 countries (Figure 3A; Table S1). Cross-validation was balanced for case-control proportions

and blocked by study in training/test folds and achieved an AUC of 0.84 (Figure 3B).

We further evaluated model performance through leave-one-study-out (LOSO) evaluation (Figures 3C and S3). Across the

27 studies, LOSO AUCs ranged from 0.7 to 0.94 (Table S5), highlighting the model's generalization capability. Low AUC values (0.59 and 0.70 in study cross-validation and LOSO evaluation, respectively) observed for some studies, e.g., the one by Hannigan et al.,⁴⁰ were likely due to low sequencing depth, as also reported in a previous meta-analysis.⁸ In other cases where low study-to-study transfer AUCs were observed (e.g., for Zhou et al.⁴¹ and Ternes et al.⁴²; Figure S3), LOSO AUCs were relatively high, suggesting diversity of datasets during classifier training can overcome technical study heterogeneity to some extent.

To gain insights into the most influential features, we also computed SHAP values for the unified CRC classifier. This analysis revealed that *Fusobacterium*, *Peptostreptococcus*, *Parvimonas*, and *Porphyromonas* were the most influential genera for CRC prediction, in line with their consistent CRC enrichment across different datasets (Figure 3D). These most influential CRC-enriched taxa were highly correlated to each other, indicating their frequent co-occurrence within the global CRC microbiome (Figure 3D).

To address the question of how taxonomic resolution affects CRC signatures, we also assessed species-level CRC enrichment patterns (on shotgun datasets using the mOTUs profiler to obtain species equivalents⁴³). Members of key CRC-enriched genera, including *Parvimonas micra*, *Peptostreptococcus stomatis*, *Fusobacterium nucleatum* subsp. *animalis*, and *Gemella morbillorum*, were among the most strongly CRC-enriched species (Figure S3; Table S6). Nevertheless, we also observed CRC-enriched species, such as *Clostridium symbiosum* and *Bacteroides fragilis*, for which the corresponding genera were not (or were only weakly) enriched in CRC. When we trained classifiers on species-level features, we observed comparable AUCs of 0.80 and 0.81 for species and genus classifiers, respectively (directly compared on WGS data only; Figure S3), suggesting that in this case higher taxonomic resolution does not directly translate to higher accuracy of microbiome-based CRC detection.

Moreover, to conceptually compare the potential diagnostic utility of microbiome profiles relative to standard non-invasive tests for CRC detection, we assessed classifier performance in studies with available fecal occult blood test (FOBT) and fecal immunochemical test (FIT) results (Figure S3). Where FOBT or FIT data were available, test performances varied widely across studies (in particular, true positive rates [TPRs] from 0.38 to 0.86), highlighting substantial heterogeneity among the different types and brands of these tests used. While fecal microbiome classifiers showed similarly variable LOSO performances across these datasets (TPR ranging between 0.22 and 0.55), particularly low microbiome-based detection rates for Hannigan et al.⁴⁰ and Baxter et al.⁴⁴ could in part be explained by their low sequencing depth (Figure 3C). While limited to a few datasets, our comparison suggests that FIT results are currently unmet by microbiome-based classifiers. Larger study populations with results of state-of-the-art FIT tests are needed to more comprehensively assess the complementarity between non-invasive clinical and microbiome-based tests and the potential of combination diagnostics.^{5,45}

Early pre-cancerous lesions are another challenge for non-invasive CRC detection. To evaluate potential fecal microbiome

associations with colorectal adenomas (AD) (as cancer precursors), we leveraged the pooled set of microbiome profiles from patients with AD ($n = 1,551$) that were part of our large data collection (see Table S1). We first compared fecal microbiome signatures between CRC, adenoma, and neoplasia-free samples using our harmonized genus-level taxonomic profiles and previously described meta-analysis approach (see Table S7 and STAR Methods). While CRC samples differed markedly and consistently from both AD and neoplasia-free controls, adenoma-associated bacterial enrichments relative to controls were substantially weaker than, and showed only minimal concordance with, CRC-associated enrichments (Figures S4A–S4D). When we applied our unified CRC classifier to AD profiles, the resulting classification scores remained below those for CRC profiles and were less well distinguished from controls (Figure S4E). This CRC-trained classifier, as well as a classifier that we specifically retrained to discriminate between AD and control profiles (see STAR Methods), both yielded highly variable AD detection accuracies across cohorts, with AUCs ranging between 0.35 and 0.70 and between 0.48 and 0.79 for CRC-trained and AD-trained classifiers, respectively (Figures S4F and S4G; Table S8). Individual genera made only modest contributions to adenoma-control classification in our models (Figure S4H; Table S8), reflecting the relatively subtle abundance changes associated with AD. Our finding of limited detectability of AD is consistent with most earlier reports (Figure S4I).

Fecal and tumor-resident microbiome signatures across CRC stages and locations

Fecal microbiome readouts are frequently criticized as an imperfect surrogate for pathophysiological conditions that are localized to certain areas of the gastrointestinal tract.^{45,46} To investigate this issue for colorectal tumors, we directly compared the tumor tissue-resident to the fecal microbiomes. We performed a meta-analysis of cancer tissue samples in comparison to adjacent non-tumor tissue (see STAR Methods) encompassing seven studies with paired samples (total $n = 906$; 457 tumor and 449 adjacent non-tumor tissues; Figure 4A; Table S9). Across study populations, tumor tissue, in comparison to non-tumor tissue, was enriched for previously described CRC-associated genera, including *Fusobacterium*, *Campylobacter_A*, *Campylobacter_B*, *Leptotrichia*, *Parvimonas*, *Peptostreptococcus*, *Gemella*, and *Porphyromonas*. By contrast, health-associated commensals such as *Faecalibacterium* and *Lachnospira* were depleted in tumor tissue (Figure 4A). Overall, the CRC tissue microbiome signature showed striking similarity to the fecal CRC signature in that *Fusobacterium*, *Peptostreptococcus*, *Parvimonas*, *Porphyromonas*, and several other genera are consistently present and significantly enriched in CRC (Figure 4B). While we found many genera to be enriched both in CRC tissue samples (relative to adjacent non-tumor tissue) and in fecal samples from CRC patients (relative to tumor-free controls), we also noted that their cumulative relative abundance was on average much higher in tissue compared with stool, plausibly as a consequence of tumor-resident microbes diluting in stool (Figure 4C).

Next, we sought to understand how key CRC-enriched bacteria were distributed across tumor stages and anatomical locations and whether this differed between tissue and fecal

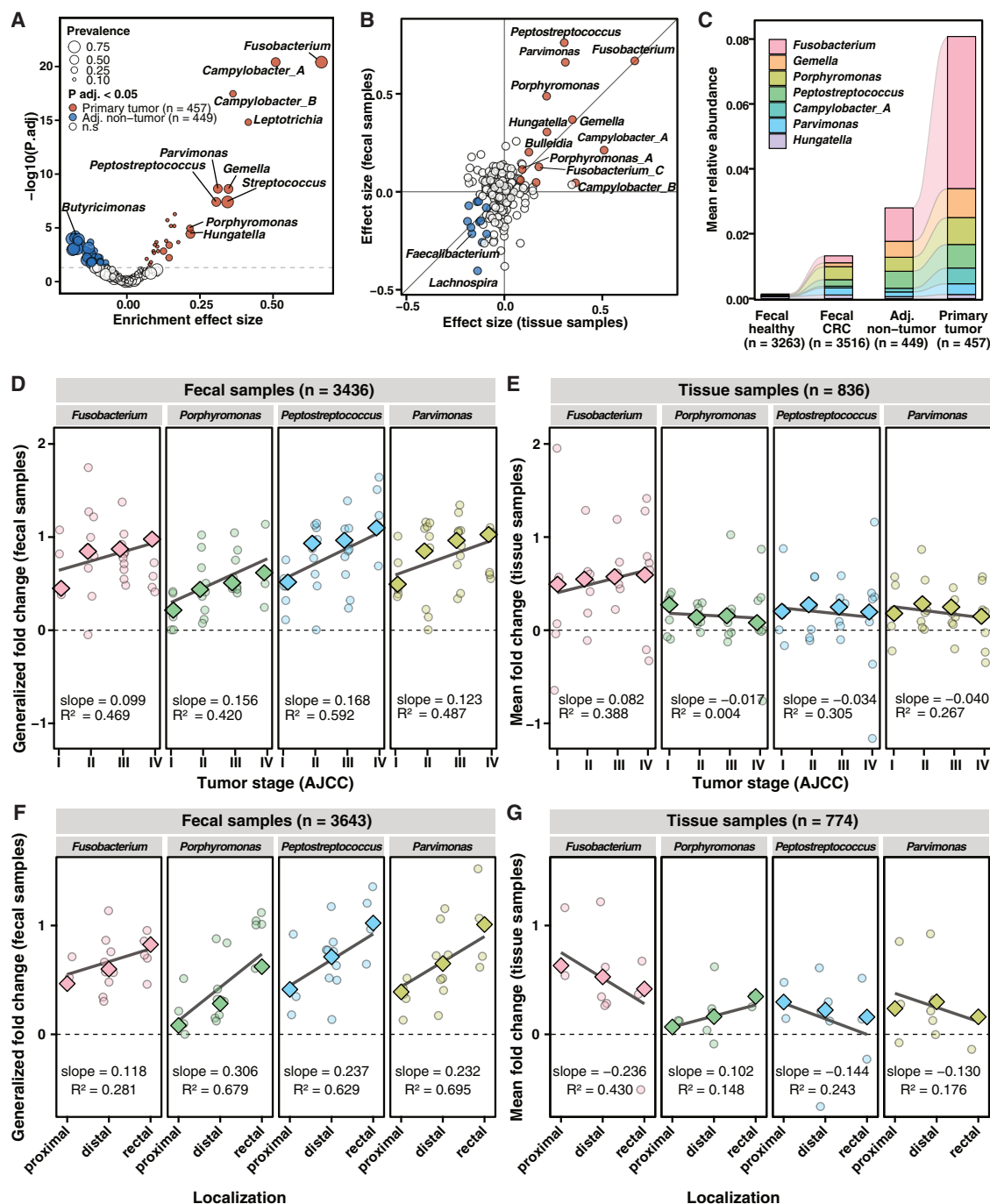


Figure 4. Fecal CRC microbiome signatures capture key tumor tissue-enriched taxa, albeit with biased representation across tumor stages and locations

(A) Meta-analysis of 295 genera enrichments in 16S profiles of CRC tumors and adjacent non-tumor tissue ($n = 906$ from 7 datasets), performed using LMM to account for study heterogeneity (see Table S9 and STAR Methods).

(B) Comparison of genus-level CRC enrichment in fecal and tumor tissue samples.

(C) Mean relative abundance of selected CRC-associated genera in fecal samples from tumor-free individuals ($n = 3,263$) and CRC patients ($n = 3,516$), and in tissue samples from adjacent non-tumor tissue ($n = 449$) and tumor tissue ($n = 457$) of CRC patients.

(D and E) Enrichment of key CRC-associated genera across tumor stages (AJCC I-IV) in fecal (D) and tissue (E) samples. Enrichment in fecal samples increases with advancing stage, whereas it remains largely stable across stages in (paired) tissue samples.

(F and G) Enrichment of key CRC-associated genera across tumor locations (proximal, distal, rectal) in fecal (F) and tissue (G) samples. While fecal microbiome profiles suggest stronger enrichment (better detectability) for rectal and distal colon tumors, this is not observed in (paired) tissue samples. Enrichment in fecal samples (D and E) is measured using the generalized fold change. In paired tissue samples, standard fold changes were averaged (see STAR Methods).

microbiome profiles. While we observed key CRC-enriched taxa to increase with advancing tumor stage in fecal profiles (Figure 4D), those same taxa rather showed a uniform enrichment across tumor stages in tissue profiles and were clearly detectable in early-stage tumors already (perhaps with the exception of *Fusobacterium*, which exhibited higher variability across studies; Figure 4E). Similarly, while fecal profiles suggested these taxa were more strongly enriched in distal and rectal tumors compared with proximal tumors (Figure 4F), tissue profiles rather showed a weak opposing trend of enrichment in proximal compared with distal and rectal tumors, possibly due to biological differences in tumor biology between anatomic locations (Figure 4G). Together, these findings suggest that current fecal microbiome profiling approaches exhibit slightly lower sensitivity for the non-invasive detection of early-stage and right-sided tumors despite these tumors being enriched in key CRC-associated bacteria.

Associations of the CRC microbiome signature with diet

While our study cannot resolve whether gut microbes causally contribute to CRC etiology, accumulating evidence from genomic and preclinical studies supports plausible mechanisms by which bacteria drive CRC development.^{4,14–19,30,31,34,42,47,48} Assuming the gut microbiome to be a risk factor for CRC, it is an intriguing question whether and how it could be modulated for CRC prevention.

To explore this idea, we examined whether the CRC microbiome signature is related to known dietary risk factors for CRC, such as fiber intake,⁴⁹ and how the relevant taxa respond to dietary interventions. We first re-analyzed cross-sectional studies, which recorded participants' habitual diets, and found estimated fiber intake (among other plant-derived dietary components; see Figure S5) to be significantly inversely correlated with the CRC signature score (Figures 5A and 5B). This was observed in both CRC patients and tumor-free individuals in the case-control study by Yachida et al.¹⁰ (Spearman's $\rho = -0.18$, $p = 0.00026$) and among baseline samples of a study with healthy overweight individuals⁵⁰ (Spearman's $\rho = -0.2$, $p = 0.013$). Similarly, individuals adhering to plant-based diets (vegetarian or vegan) exhibited significantly lower CRC signature scores compared with omnivorous individuals⁵¹ (Wilcoxon rank-sum test, $p = 0.043$; Figure 5C). Likewise, baseline samples from the study by Healey et al.⁵² also showed lower CRC microbiome signature scores for those individuals consuming a high-fiber diet compared with those on a low-fiber diet (Wilcoxon rank-sum test, $p = 0.043$; Figure S5).

When re-analyzing dietary intervention studies that aimed to increase fiber intake, we found the CRC microbiome signature scores to decrease significantly across several types of interventions (Figures 5D–5G and S5). This was observed in studies that contrasted periods of a Westernized diet (low in fiber and high in saturated fats) with high-fiber interventions,^{53,54} and among the 10 most influential taxa on the CRC classifier (see Figure 3D), those genera that were negatively associated with CRC most strongly responded (Figure S5). In similar re-analyses of published studies, we also observed lowered CRC signature scores following a placebo-controlled fiber intervention⁵² (exclusively after fiber supplementation, not placebo; see Figures 5E and S5) and in response to consumption of pea-fiber-containing

snacks^{54,55} (Figures 5F, 5G, and S5). Altogether, these results support the concept that the abundance of CRC-associated gut microbes is associated with habitual fiber intake and amenable to modulation by dietary fiber supplementation.

Functional metagenome analyses reveal CRC-enriched metabolic pathways and VFs

To investigate functional microbiome alterations in CRC, we compared CRC and control metagenomes ($n = 1,948$ WGS profiles from 14 studies) for differentially abundant gut metabolic modules (GMMs),⁵⁶ as well as Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways and modules,⁵⁷ using LMMs to control for study heterogeneity (see STAR Methods). This corroborated a previously reported shift in CRC microbiome functional repertoires toward amino acid degradation^{5,9}; consistently across GMM and KEGG, many of the most significant and most strongly CRC-enriched modules were involved in amino acid degradation (Table S10). Furthermore, we noted modules related to redox and respiratory systems to be more abundant in CRC metagenomes (Table S10). This increased abundance of genes involved in aerobic respiration could be related to a higher abundance of aerotolerant Gram-negatives (e.g., *Porphyromonas*, *Escherichia*, and *Morganella*) thriving in the CRC microenvironment, which is characterized by higher levels of reactive oxygen species.⁵⁸ Finally, among non-metabolic KEGG modules, we detected many genes encoding components of type III, IV, and VI secretion systems to be enriched in CRC metagenomes (Figure S6). These secretion systems allow pathogens to deliver effector proteins and toxins into host cells.^{59,60} Together with the elevated abundance of lipopolysaccharide biosynthesis ($p = 7.2e-12$), these results suggest an expansion of pathobionts and virulence mechanisms in CRC.

For a more comprehensive characterization of gut microbial VF repertoires, we combined Virulence Factor Database (VFDB) 2.0 entries⁶¹ with manually curated VF annotations (Tables S11 and S12) to increase the coverage of VFs of relevance in CRC (Figure S6; see STAR Methods). Differentially abundant VFs included strongly CRC-enriched adhesins (fap2, pcwbr2, fadA, fliA, srtC-1/srtB, cmpA, srtC4, and fomA), oxidative stress-related genes (msrA/B) (Figure 6A; Table S13), and dps (FN1079), which encodes a fusobacterial VF implicated in tumor progression.⁶² Another notable CRC-enriched gene is pcwbr2, which encodes a *Peptostreptococcus anaerobius* surface protein mediating tumor-cell adhesion via integrin $\alpha 2/\beta 1$ and pro-tumorigenic signaling.³⁰

Genomic and biogeographic analysis of *Fusobacterium* strains and VFs

To further resolve and contextualize fusobacterial VFs, we performed a strain-level phylogenomic analysis of the *Fusobacterium* genus using a total of 698 genomes, which revealed intriguing lineage-specific gene presence/absence patterns: CRC-enriched fadA,⁴⁸ fap2,¹⁴ fliA,⁶⁴ fusolisins,⁶⁵ and msrA/B⁶⁶ genes were largely restricted to genomes and species of the monophyletic *F. nucleatum*/*F. periodonticum* subclade (Figure 6B). This VF survey further highlights the *Fusobacterium* subclade in which the most strongly CRC-associated fusobacterial VFs and species coalesce.

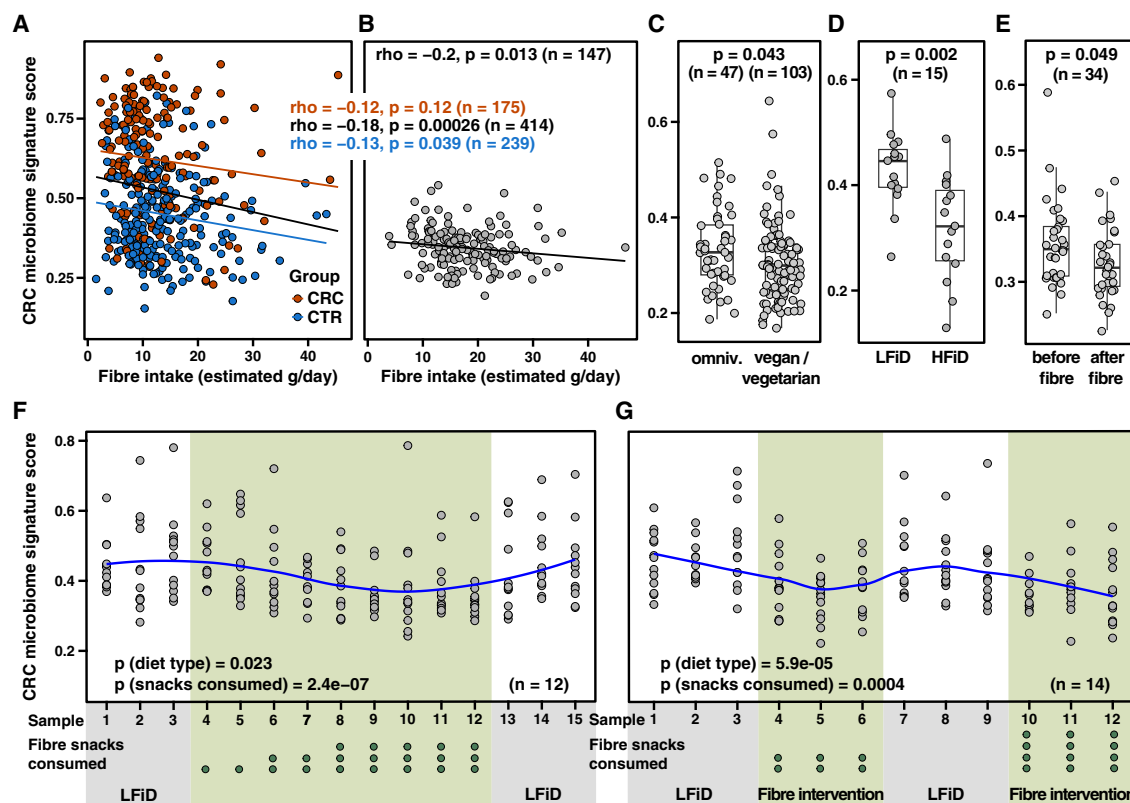


Figure 5. The CRC microbiome signature score is inversely associated with dietary fiber intake and can be modulated by dietary interventions

(A) Correlation between estimated daily fiber intake and the CRC microbiome signature score in a CRC case-control study¹⁰ broken down by group: CRC patients (red), controls (blue), and both combined (black). Spearman's correlation coefficients and *p* values are indicated.

(B) Same correlation assessed independently in healthy (tumor-free) overweight individuals (cross-sectional analysis at baseline).⁵⁰

(C) Differences in CRC microbiome signature score across distinct habitual diet groups from a third study comparing omnivorous with vegan and vegetarian individuals.⁵¹

(D) CRC microbiome signature scores are significantly reduced in healthy individuals switching from a Westernized (LFID, low-fiber diet) to a fiber-enriched Mediterranean diet (HFID, high-fiber diet).⁵³

(E) CRC microbiome signature scores compared in healthy participants before and after a controlled fiber supplementation intervention.⁵²

(F and G) Consumption of fiber-rich snacks leads to decreasing CRC microbiome signature scores in two human intervention studies (both reported in Delannoy-Bruno et al.⁵⁴). In both studies, overweight or obese participants followed a controlled LFID. Introduction of fiber-rich snacks (periods highlighted in green corresponding to days 15 to 35 in the first study, F, and to days 12 to 25 as well as days 36 to 49 in the second study, G) resulted in significant reduction of CRC microbiome signature scores as assessed by LMMs in which either diet types (LFID versus fiber intervention) or the number of snacks consumed (green dots at the bottom) were modeled as fixed effects.

With our comprehensive dataset, we next sought to re-examine cancer enrichment and biogeography of *Fusobacterium* lineages beyond previous work.^{63,67,68} A previous study reported two distinct *F. nucleatum* subspecies *animalis* subclades—termed “Fna C1” and “Fna C2”—with differential virulence potential and enrichment in CRC tissues.⁶³ While our phylogenomic analysis supports the delineation of these clades (Figure 6B), it indicates that they correspond to pre-existing species- or subspecies-level clades: while molecular delineation by mOTUs⁴³ suggests species, NCBI annotations are ambiguous as to species or subspecies assignment. Fna C2 corresponds directly to mOTU 01001, annotated as *F. nucleatum* subspecies *animalis*, or *F. animalis*, according to NCBI taxonomy (this and other fusobacterial mOTUs also contain *F. nucleatum* NCBI annotations that are not further resolved to subspecies). Fna C1, on the other hand, corresponds

to mOTU 01002, annotated as *F. nucleatum* subspecies *vincentii/watanabei* (with agreeing designations in NCBI). An independent taxonomic re-evaluation of Fna C1 and Fna C2 independently proposed a reclassification consistent with ours.⁶⁹

We finally investigated CRC enrichment of *Fusobacterium* spp. across the geographically diverse patient populations in our meta-analysis. While we found *F. nucleatum* subsp. *animalis* (mOTU 01001, Fna C2) to be uniformly CRC-enriched across all continents, other subspecies exhibited more heterogeneity across continents. Within the *F. nucleatum* clade, *F. n. subspecies vincentii/watanabei* (mOTU 01002, Fna C1), *F. n. subspecies nucleatum* (mOTU 01003), and *F. n. subspecies polymorphum/hwasookii* (mOTU 01004) all showed little to no enrichment in US patient populations but stronger CRC associations in Asian populations (Figure 6C). Even more pronounced heterogeneity with peaked CRC enrichment in Asian populations was observed

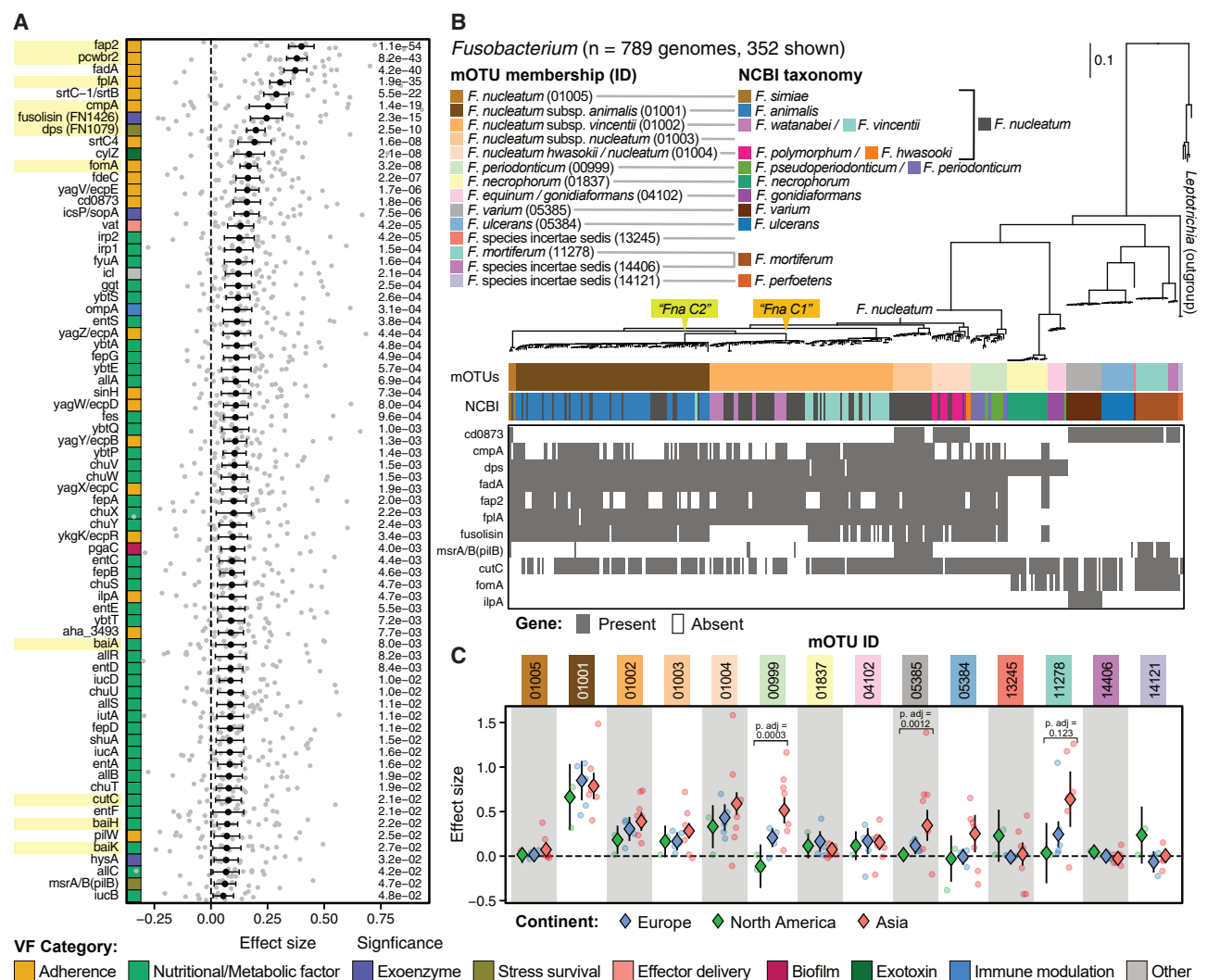


Figure 6. Functional meta-analysis of VFs and *Fusobacterium* (sub)species

(A) CRC enrichment of VFs in WGS data (see Tables S11–S13). Large, black dots correspond to the overall enrichment effect size estimate from meta-analysis, with 95% confidence intervals shown by lines, and smaller, gray dots indicate study-specific estimates. VFs highlighted in yellow denote manually curated additions not represented in VFDB.

(B) Core-gene phylogenetic tree and heatmap showing the presence and genomic distribution of VFs across 14 human gut *Fusobacterium* mOTUs, labeled by both mOTU membership and NCBI taxonomy ID. The Fna C1 and Fna C2 labels correspond to the two *F. nucleatum* clades proposed in Zepeda-Rivera et al.⁶³ (see main text). For all mOTUs except *F. nucleatum* subsp. *animalis* (01001) and *F. nucleatum* subsp. *vincentii* (01002), we show at most 20 genomes per mOTU.

(C) Dot plot reflecting enrichment of *Fusobacterium* mOTUs in CRC patient populations stratified by continent (effect size estimates of continent-stratified meta-analyses shown as diamonds with 95% confidence intervals indicated by lines). Significance labels denote between-continent heterogeneity assessed by a subgroup Q-test (χ^2 , df = 2) with Benjamini-Hochberg correction. All non-displayed adjusted *p* values were >0.12.

for *F. periodonticum*, *F. varium*, and *F. mortiferum* (Figure 6C), consistent with and substantially extending prior reports⁶⁷ (on Chinese populations). Together, these results support the notion that while CRC-associated bacterial genera are globally consistent, pathogenic species and strains nevertheless can display intriguing biogeographic patterning across the global human population.

DISCUSSION

This study, to our knowledge, represents one of the largest single-disease microbiome meta-analyses based on consistent re-

profiling to date, encompassing a total of 6,779 fecal and 906 intestinal tissue samples from 34 study populations. Beyond its scale, a key strength of this work lies in the methodological advance toward standardization and harmonization of microbial profiles. By combining consistently profiled 16S and WGS sequencing data for association testing and ML modeling (see Figure 1), we were able to substantially expand upon previous meta-analyses by evaluating taxonomic CRC microbiome signatures across a wide range of studies, geographic regions, sequencing approaches, and subgroups of CRC, even though major human populations in Africa, Asia, and South America remain undersampled.

A major aim of our study was to re-examine the clinically important question of whether EO-CRC-associated gut microbiome signatures substantially differ from those of LO-CRC.^{22,24,25} Yang et al. and Kong et al. reported significant taxonomic and functional differences between EO-CRC and LO-CRC; however, they did not perform direct cross-study validation, and their patient populations were of limited size and geographically confined to China. By contrast, Qin et al. recently analyzed a larger dataset from multiple centers (also in China) and concluded that microbial features were largely shared across age groups.²⁵ In our meta-analysis, which included 22 age-annotated study populations from around the world, offering a much more comprehensive view, we found little evidence for fecal CRC microbiome signatures differing by age of onset (see Figure 2). Instead, our results strongly support the concept of a uniform CRC signature: key CRC-enriched genera, most importantly *Peptostreptococcus*, *Fusobacterium*, *Parvimonas*, and *Porphyromonas*, showed equal increases in EO-CRC and LO-CRC, varied only minimally by age of onset, and were instrumental predictors in unified ML models that generalized well across age groups (see Figures 3 and 4).

Performing analysis of CRC subgroups delineated by tumor stage and location, we found key bacterial genera to be enriched in fecal readouts across all stages and locations, albeit more weakly so for early-stage and right-sided or proximal CRC (see Figure 4). In line with lower detection rates for early-stage CRC, AD proved difficult to detect, both when using the universal CRC classifier or an adenoma-specific one (see Figure S4). Together with earlier studies showing low adenoma detection rates,^{8,21} our results suggest that the field currently lacks reproducible and generalizable fecal microbiome markers for AD. These findings prompted the question of whether microbial colonization may occur only late in tumor progression—suggestive of a bacterial “passenger” model. To rule out indirect effects in fecal surrogate readouts, we directly analyzed tumor tissues for enrichment of key CRC-associated bacteria, stratified by stage and location. Intriguingly, in contrast to fecal patterns, bacterial tumor tissue enrichment (relative to adjacent non-tumor tissue) was largely uniform across stages and locations (see Figure 4), consistent with a “driver” model of early microbial tumor colonization. We propose that this observed discrepancy between tissue and fecal profiles likely reflects a “diluted” readout of tumor-colonizing bacteria in feces, hampering detection of characteristic CRC-associated bacteria for tumors that are more localized (early stage) or farther apart from the rectum (i.e., right-sided, proximal CRC). More sensitive fecal microbiome readouts (e.g., higher sequencing depth) might improve non-invasive microbiome-based detection of early-stage and right-sided CRC.

With CRC microbiome signatures firmly established by this and earlier works,^{5–10} the questions of how CRC-associated bacteria (perhaps emerging first in the oral microbiome^{8,14,70}) colonize the colon and how this process may be reversed become increasingly relevant—not least because they are fundamental to developing potential microbiome modulation strategies for cancer prevention. In exploring this idea on the basis of our unified CRC microbiome scoring models, we documented a significant inverse relation between dietary fiber intake, for which strong epidemiological evidence supports a cancer-protective role,⁴⁹ and the CRC microbiome score, both in cancer patients and tumor-free individuals

(see Figure 5). More importantly, we also showed that several interventions aimed at increasing dietary fiber content effectively lowered CRC microbiome scores. These interventions tended to increase genera, such as *Agathobaculum* and *Lachnospira* (see Figure S5), which appear to be credible antagonists of CRC-enriched bacteria, suggested by their inverse influence on the CRC microbiome score and their anticorrelated abundance with CRC-enriched taxa (see Figure 3D). While our results provide a proof of concept, we cannot fully resolve complex dietary effects, making it possible that other components of plant-based diets, beyond fiber, have contributed to the observed relationship (see Figure S5). More work, including prospective studies, is needed to delineate dietary components with efficacy in reducing CRC risk and how to rationally identify and promote the growth of those gut microbes that most effectively antagonize CRC-enriched pathobionts.

Our investigation of CRC-associated bacterial strains, genes, and pathways (see Figure 6) revealed *Fusobacterium* virulence genes such as *fadA*, *fap2*, *fplA*, *cmpA*, *fomA*, and *fusolisin*—all known to promote bacterial adhesion, modulate antitumor immune response, or aid in epithelial invasion—to be concentrated within the *F. nucleatum* clade. While *F. nucleatum animalis* from within this clade exhibited a globally consistent pattern of CRC enrichment, other (sub-)species, including *F. periodonticum*, *F. varium*, and to some extent also *F. mortiferum*, showed CRC enrichment patterns that differed between continents. The broader diversity of CRC-associated fusobacterial (sub-)species in Asian populations warrants further investigation into biogeographic differences of strain-specific virulence mechanisms, reservoirs, and transmission routes. While these results highlight geographic stratification at the (sub-)species level, our work at the same time strongly supports the existence of bacterial genera that are universally CRC-enriched across the globe, both in tumor tissue and fecal microbiome profiles.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Georg Zeller (georg.zeller@gmail.com).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- All data used in this meta-analysis are available from public archives (see Table S1 for dataset identifiers).
- All statistical and ML results generated from this study are provided as supplemental tables.
- To further improve the reusability of the data, all taxonomic and functional microbiome profiles used for this study, as well as the scripts used for analysis and figure generation, have been deposited (see [key resources table](#)) and are publicly available as of the date of publication. The code has also been deposited at GitHub: <https://github.com/zellerlab/crc-meta-ii>.
- Any additional information required to re-analyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.P., N.K., and G.Z.; data curation, S.P., N.K., F.S., S.R., and A.L.; taxonomic and functional profiling, S.P., F.S., Q.R.D., and A.L.; software development, S.P., N.K., and M.E.; investigation, S.P., N.K., M.E., F.S., S.R., Q.R.D., and A.L.; visualization, S.P., N.K., M.E., and F.S.; writing – original draft, S.P., N.K., M.E., and G.Z.; writing – review and editing, S.P., N.K., M.E., S.Z., M.Z., and G.Z.; funding acquisition, S.Z., M.Z., and G.Z.; resources, F.S., S.R., and M.L.; and supervision, G.Z.

DECLARATION OF INTERESTS

G.Z. is a named inventor on a patent (EP2955232A1: Method for diagnosing adenomas and/or CRC based on analyzing the gut microbiome).

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used large language models to improve clarity and readability. After using this tool or service, the authors reviewed and edited the content as needed, taking full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Raw WGS and 16S fecal microbiome sequencing data	ENA/SRA/DBJ	See Table S1 for individual BioProject accessions
Uniformly processed taxonomic and functional profiles with sample metadata	This paper	https://doi.org/10.5281/zenodo.20310629
Source code	This paper	https://doi.org/10.5281/zenodo.20309766
Software and algorithms		
R (v 4.3.3)	R Core Team	https://www.r-project.org/
BBDuk (v38.91)	Bushnell ⁷¹	https://sourceforge.net/projects/bbmap/
IDTAXA (DECIPHER package v3.0.0)	Murali et al. ⁷²	https://doi.org/10.1186/s40168-018-0521-5
mOTUs (v3.1)	Ruscheweyh et al. ⁴³	https://github.com/motu-tool/mOTUs
mOTUs extender (v3.1)	Ruscheweyh et al. ⁴³	https://github.com/motu-tool/mOTUs-extender
SIAMCAT R package (v2.7.2)	Wirbel et al. ²⁶	https://github.com/zellerlab/SIAMCAT
ranger R package (v0.16.0)	Wright and Ziegler ⁷³	https://github.com/imbs-hl/ranger
kernelshap R package v0.7.0	Lundberg and Lee ⁷⁴	https://github.com/ModelOriented/kernelshap
pROC R package (v1.18.5)	Robin et al. ⁷⁵	https://doi.org/10.1186/1471-2105-12-77
lme4 R package (v4.1.1-35.5)	Bates et al. ⁷⁶	https://github.com/lme4/lme4
partR2 R package (v0.9.2)	Stoffel et al. ⁷⁷	https://github.com/mastoffel/partR2
vegan R package (v2.6-6.1)	Oksanen et al. ⁷⁸	https://github.com/vegandevs/vegan
labdsv R package (v2.1.0)	Roberts et al. ⁷⁹	https://github.com/cran/labdsv
coin R package (v1.4-3)	Hothorn et al. ⁸⁰	https://github.com/cran/coin
omixerRpm (v0.3.3)	Darzi et al. ⁸¹	https://github.com/omixer/omixer-rpmR
gffquant (v2.10)	NA	https://github.com/csChu/gff_quantifier
BWA-MEM (v0.7.17)	Li et al. ⁸²	https://github.com/lh3/bwa
NGLess / NG-meta-profiler	Coelho et al. ⁸³	https://github.com/ngless-toolkit/ngless
DIAMOND (v2.0.13)	Buchfink et al. ⁸⁴	https://github.com/bbuchfink/diamond
PyHMMER (v0.11.0)	Larralde and Zeller ⁸⁵	https://github.com/althonos/pyhmmmer
fetchMGs (v2.0.1)	Ruscheweyh et al. ⁴³	https://github.com/motu-tool/fetchMGs
MAFFT (v7.505)	Katoh et al. ⁸⁶	https://doi.org/10.1093/nar/gkf436
FastTree (v2.1.10)	Price et al. ⁸⁷	https://doi.org/10.1371/journal.pone.0009490
BLASTp	Altschul et al. ⁸⁸	https://doi.org/10.1016/S0022-2836(05)80360-2
KEGG Mapper	Kanehisa et al. ⁸⁹	https://doi.org/10.1002/pro.4172
KOFAMKoala prokaryotic HMMs	Aramaki et al. ⁹⁰	https://doi.org/10.1093/bioinformatics/btz859

METHOD DETAILS

Study inclusion and data curation

In order to systematically identify relevant studies for inclusion in this meta-analysis, we performed a comprehensive, keyword-based PubMed search followed by a series of filtering steps. The initial PubMed search string was as follows and applied to all abstracts in the PubMed database, yielding a total of 956 articles as of January 2023.

((colon[tiab] OR colorectal[tiab] OR rectal[tiab] OR bowel[tiab]) AND (cancer*[tiab] OR precancer*[tiab] OR carcino*[tiab] OR polyp[tiab] OR adenoma*[tiab] OR neoplas*[tiab] OR malign*[tiab])) AND (shotgun[tiab] OR metagenom*[tiab] OR (meta[tiab] AND sequencing[tiab]) OR WGS[tiab] OR 16S[tiab] OR amplicon[tiab] OR “ribosomal sequencing”[tiab] OR “associated microb”[tiab] OR “associated bacteri”[tiab] OR (profile*[tiab] AND microb*[tiab]) OR “bacterial tax”[tiab]) AND ((human*[tiab] OR patient*[tiab] OR participant*[tiab] OR subject*[tiab] OR cohort*[tiab] OR individual*[tiab] OR “tumor/normal pair”[tiab]) NOT (“Case Reports”[publication type]) NOT (“Review”[publication type]) AND (2011:2023[dp])).

We then filtered the studies with the following criteria: Study needs to include novel stool sequencing data, study design needs to be appropriate (case/control design including neoplasm and control data), participants must not be pre-treated with antibiotics or chemotherapy and must not have had colonic surgery. Furthermore, we required a minimum total of 30 samples and at least 15 cases. Finally, sequencing data and metadata needed to be accessible through ENA/SRA or DDBJ. These filtering steps resulted in a total of 27 datasets for final inclusion in this work (see [Table S1](#)). Study set filtering was independently performed by NK and FS, resulting in the same set of final studies for inclusion.

For the CRC tissue meta-analysis, we compiled seven publicly available 16S rRNA amplicon datasets of CRC tissue samples for which information on tumor stage was available. Only studies including at least 15 CRC tumor and adjacent non-tumor tissue profiles were considered. The datasets incorporated in this analysis were retrieved through the NCBI and ENA repositories using their corresponding BioProject accessions (see [Table S1](#)).

Preprocessing of WGS and 16S data

Raw sequencing reads from WGS and 16S data were processed using *bbduk* (v38.91).⁷¹ For WGS data, reads underwent several preprocessing steps to remove low-quality sequences: Low-quality bases were trimmed from both ends (*qtrim* = *rl*, *trimq* = 3), followed by the removal of entire reads falling below a quality threshold (*maq* = 25). Adapter sequences were then removed (*ktrim* = *r*, *k* = 23, *mink* = 11, *hdist* = 1, *tpe* = *true*, *tbo* = *true*) and reads shorter than 75 base pairs were discarded. For 16S data, reads underwent low-quality base trimming (*qtrim* = *rl*, *trimq* = 10), followed by targeted removal of 5' amplicon primers from R1 and any potential reverse complement primers from R2 (*ktrim* = *l*, *k* = 9, *mink* = 1, *hdist* = 1, *restrictleft* = 50, *maq* = 20). Similarly, 3' primers were removed from R2, along with any potential reverse complement sequences from R1 (*ktrim* = *r*, *k* = 9, *mink* = 1, *hdist* = 1, *restrictright* = 50, *maq* = 20). Finally, reads shorter than 75 base pairs were discarded. Tissue- and stool-derived 16S data was preprocessed in the same fashion.

Taxonomic Profiling

For profiling of tissue and stool derived 16S data, we used *IDTAXA*⁷² from the *DECIPHER* package (v3.0.0) with parameters *strand* = both and *threshold* = 40 and the GTDB r207 (modified) training set. WGS data were profiled using *mOTUs* (v3.1)⁴³ with default settings. Results were mapped to GTDB v207 using a mapping to GTDB v207 provided by the authors of *mOTUs3* (<https://zenodo.org/records/10275750>). From both 16S and WGS data, we removed samples with fewer than 1000 total microbial counts from further analysis.

Functional profiling

To compute functional profiles from WGS data, we employed *gffquant* (v2.10) (https://github.com/csChu/gff_quantifier) in conjunction with a reduced version of the Global Microbial Gene Catalogue (GMGC) human gut nr95 catalog⁴³. This refinement involved excluding genes present in fewer than 0.5% of the shotgun metagenomes used to construct the GMGC, yielding a final catalog of 13,788,251 distinct, non-redundant genes.

Reads were mapped to the reduced human gut catalog using *BWA-MEM* (v0.7.17)⁸² with default settings. Only alignments longer than 45 bases and at least 97% sequence identity to the reference were retained. Reads aligning to multiple genes were counted fractionally across all matching genes. Alignment counts were normalized based on gene length, then further scaled following the method implemented in *NGLess*.⁸³ The final relative abundance values were computed by normalizing against the total number of mapped reads that met the filtration criteria.

To obtain virulence factor relative abundances, we mapped the final set of virulence factor protein sequences (see section below on curation of virulence factors) to the GMGC using *DIAMOND* (v2.0.13).⁸⁴ The results were then filtered using a minimum sequence identity of 50% along with query and subject coverage filters of at least 75%. For GMGC proteins with multiple virulence factor hits, the VF hit with the lowest E-value was retained. In addition to the final set of virulence factors, we included bile acid-inducible (*bai*) genes. Hidden Markov Models (HMMs) for 11 *bai* genes were obtained from a previous study⁹ and used to search the GMGC gene catalog using *PyHMMER*⁸⁵ (v0.11.0). For each *bai* gene, matching GMGC sequences were aligned to reference sequences from *Clostridium hylemonae* and *Clostridium scindens* using the Smith–Waterman algorithm. Only high-confidence annotations with a minimum protein sequence identity of 80% across the alignment were retained. The relative abundances of GMGC genes were then aggregated at the virulence factor level (or *bai* gene) to generate functional profiles for downstream analyses.

To derive KEGG orthologous group (KO) functional profiles, we used KOFAMKoala prokaryotic HMMs to annotate the GMGC. KEGG pathways and modules were further refined, keeping only those with at least 50% and 60% prokaryotic KOs, respectively. Gut metabolic modules (GMMs) were inferred based on KOs using the R package *omixerRpm* (v0.3.3)⁸¹ with default parameters and a pathway coverage threshold of 0.5. We then used the KEGG Mapper portal⁸⁹ to map the differentially abundant KOs onto KEGG pathway maps and verify which pathways are involved in the context of CRC.

Curation of virulence factors

We obtained virulence factor nucleotide sequences from the expanded Virulence Factor Database (VFDB 2.0).⁶¹ VFDB entries were translated into amino acid sequences using the bacterial/archaeal genetic code (NCBI translation [Table S11](#)) as implemented in the

Biostrings R package (v2.72.1). Sequences containing ambiguous nucleotide characters were excluded before translation. Protein sequences were quality filtered to remove internal stop codons, trimmed for terminal stop codons, and collapsed into a non-redundant *Fusobacterium* VF reference dataset.

Additionally, we conducted a manual literature search to identify additional virulence factors (VFs) associated with colorectal cancer (Table S11). To search for additional close homologs of literature-described VFs (Table S12), we performed BLASTp⁸⁸ searches for each manually added VFs using the Non-redundant (NR) protein sequence database, restricted to the bacterial sequences (taxid:2). The results of these BLASTp searches were manually curated (using annotation information from NCBI GenBank) to ensure compatibility between manually identified and VFDB-sourced VFs in downstream analysis. We combined manually found VFs and their putative homologs with the ones from VFDB 2.0. Finally, we discarded the VFs which were well-represented in KEGG,⁵⁷ including those related to the bacterial secretion system, capsule, chaperonins, LPS, LOS, flagella, and fimbriae, from downstream analysis (see Figure S6), as our KEGG analysis confirmed their strong representation and they are well-documented in the literature (Table S10).

Fusobacterium phylogeny and taxonomy

We obtained all *Fusobacterium* genomes from NCBI as of September 2025 ($n=995$). We cross-referenced this via NCBI assemble-IDs with Zepeda-Rivera et al.⁶³ and were able to retrieve 162/166 of the genomes they reported (in their Table S1). We furthermore included 20 random *Leptotrichia* spp. genomes from NCBI as an outgroup for phylogenetic reconstruction.

In order to assign genomes to mOTUs, we applied the mOTUs extender software (v3.1, <https://github.com/motu-tool/mOTUs-extender>) to all *Fusobacterium* genomes. We subsequently removed all genomes that could not be assigned to a mOTU and that did not contain all marker genes used by mOTUs extender. This way, we removed 297 genomes, retaining 698 genomes for further analysis.

For phylogenetic tree reconstruction, we used fetchMGs (v2.0.1, <https://github.com/motu-tool/fetchMGs>) to retrieve 40 universal phylogenetic marker genes, which we subsequently aligned using mafft⁹¹ (v7.505). We then approximated a maximum likelihood phylogeny from the concatenated marker gene alignment using FastTree⁸⁷ (v2.1.10) with parameter -gtr (Figure 6).

NCBI species identity of genomes was obtained directly from the taxonomic information linked to genomes on NCBI. mOTUs taxonomy was obtained by running the mOTUs extender as described above. Fna clade information as reported by Zepeda-Rivera et al. was obtained via mapping the NCBI assembly-IDs to Table S1 in Zepeda-Rivera et al.

VF annotation of Fusobacterium genomes

In the same set of *Fusobacterium* genomes used for phylogenetic reconstruction, we also annotated VF gene presence. To that end we leveraged the curated set of *Fusobacterium* virulence factors we assembled for this work (see section Curation of virulence factors) and followed the same procedure that we used to annotate the GMGC: We mapped the virulence factor protein sequences against proteins encoded in *fusobacterium* genomes using DIAMOND (v2.0.13).⁸⁴ The results were then filtered using a minimum sequence identity of 50% along with query and subject coverage filters of at least 75%. For *fusobacterium* proteins with multiple virulence factor hits, the VF hit with the lowest E-value was retained.

QUANTIFICATION AND STATISTICAL ANALYSIS

Evaluating microbial alpha- and beta diversity

Genus richness and Shannon diversity were calculated on rarefied counts using the *rrarefy* function from the *vegan* package⁷⁸ (v2.6-6.1) in R (v4.3.3). All samples were rarefied to a depth of 1000 counts. Comparison of Shannon diversity and genus richness between EO- and LO-CRC was performed using a blocked Wilcoxon test implemented in the R *coin* package⁸⁰ (v1.4-3) with the 'Study' variable used as a blocking factor to account for study heterogeneity.

Pairwise sample distances were calculated on features with at least 10% prevalence. We used Bray-Curtis and Canberra dissimilarities as input for all distance- and ordination-based comparisons. Dimensionality reduction was performed using the *pco* function from the *labdsv* R package⁷⁹ (v2.1.0), extracting the first two principal coordinates and calculating the variance these explained. To compute the proportion of compositional variance explained by the dataset over all microbial profiles, we computed PERMANOVAs on pairwise microbiome distances using the *adonis2* function from *vegan*.

Linear model-based meta-analysis

To identify disease-associated taxa, virulence factors, or functional features, we applied linear mixed effects models (LMMs) in a univariate manner. For all LMM models, we applied prevalence filtering to retain only features present in 10% of all samples. Afterwards, the non-rarefied relative abundance data were log-transformed using $1e-05$ as a pseudo-count before z-score standardization was applied.

For each feature, we modeled its abundance with disease status as a fixed effect and study as a random effect:

$$\text{Feature} \sim \text{Condition} + (1 \mid \text{Study})$$

The model was fitted using the `lmer` function from the *lme4* package⁷⁶ (v4.1.1-35.5). To assess the consistency of the association across studies, we also estimated study-specific effect sizes by the coefficient of a dataset-specific linear model (`lmer`). To this end, we fit dataset-wise simple linear models. All p-values were adjusted for multiple testing using the Benjamini-Hochberg (FDR) correction.

This analysis was carried out separately for comparison between CRC and CTR in both WGS and 16S taxonomic profiles at the genus level, with WGS profiles additionally analyzed at the mOTU level. Additionally we stratified by continent of study origin to perform subgroup analysis using the same approach per each continent. We additionally performed genus level comparisons for EO-CRC versus EO-CTR using different age cut-offs (<40, <45, <50 (which is the standard clinical cut-off for EO-CRC) and <55), LO-CRC versus LO-CTR, and adenoma versus CTR and adenoma versus CRC samples. In addition, we applied the same approach to KEGG orthologous groups (KOs), KEGG pathways, KEGG modules, gut metabolic modules (GMMs), and virulence factors.

To analyze CRC tissue microbiomes, we extended this approach to compare taxonomic profiles between paired CRC tumor tissue and adjacent non-tumor tissue. For each taxon, abundance was modeled with tissue type as a fixed effect and both study and patient ID as random effects, accounting for inter-study variability and paired sampling within patients.

$$\text{Taxon} \sim \text{Tissue_Type} + (1 \mid \text{Study}) + (1 \mid \text{Patient ID})$$

Generalized fold change (gFC) calculation

To quantify effect sizes across tumor stage and localization for fecal samples, we used the generalized fold change (gFC), as previously described.⁹ gFC is defined as the mean difference between case and control groups across predefined quantiles of the log-transformed abundance distributions. Quantiles from 0.1 to 0.9 (in increments of 0.1) were used.

Variance partitioning with linear models

To quantify sources of biological and technical variation on microbial profiles in specific subgroups, we fitted a separate mixed model for each covariate of interest, x , in the form of $\text{Taxon} \sim x + (1 \mid \text{Study})$, where taxonomic relative abundances were as described above.

For each model, we used the *partR2* package (v0.9.2) to estimate the variance explained for each covariate-taxon pair.⁷⁷ The *partR2* function iteratively drops fixed terms, refits the model, and returns the decrease in marginal R^2 caused by omitting that term. Marginal R^2 estimates the proportion of variance that the fixed effects explain after controlling for random intercepts (i.e. study-to-study differences), which was fully attributable to x in our simple models.

To contrast disease status and age of diagnosis, we modeled genus-level profiles that met our prevalence and abundance filters (see above) from $n = 5,663$ samples (i.e. those from the unified classifier with age information available) across each variable. We coupled these estimates with iterative nested likelihood ratio tests in a model that contained both disease and age status to confirm the relative statistical significance of these terms, after FDR correction with Benjamini-Hochberg (as in Figure 2).

Machine learning validation strategies

We fit and evaluated the machine learning models in this work using various resampling and data partitioning strategies. We applied random subsampling to both training and testing sets, such that the proportion of cases and controls are equal in both sets for all resampling strategies and for all comparisons, with the exception of the adenoma classifier where we used all data without subsampling (Figure S4). The following validation strategies aimed at assessing different aspects of model generalization, either within a study (or group of studies), across studies, sequencing technologies and across age groups.

Cross-validation

Whenever models were trained and evaluated using cross-validation, we used 10 times resampled 10-fold cross-validation with study information blocked such that samples from a study would either all be in the training or the testing set. Final prediction probabilities were defined as the median prediction probabilities over the resampled models.

Validation across sequencing methods

In order to understand if CRC microbial signatures were similar across sequencing technologies, we used the cross-validated models trained on 16S CRC/control data and evaluated their classification performance on WGS sequencing data and vice versa.

Cross-age validation

Similarly, in order to understand if CRC microbial signatures were similar across age of onset, we used the cross-validated models trained on EO-CRC/control data and evaluated their classification performance on LO-CRC and age-appropriate control samples and vice versa.

Study-to-study transfer and Leave-one-study-out (LOSO) validation

After establishing that microbial signatures were similar across sequencing methods and age of onset, we sought to understand if we could pool data across all studies to train a unified CRC classifier. To this end, we evaluated study-to-study variability in CRC microbial signatures by employing study-to-study transfer as well as leave-one-study-out (LOSO) validation. For study-to-study transfer, we trained classifiers on all CRC/control samples of one dataset and evaluated its performance on all CRC/control samples of the held out datasets. For LOSO validation, we trained classifiers on all CRC/control samples of all dataset except one and evaluated classifier performance on the held-out dataset (repeating this process for each dataset^{8,9}).

Unified classifier

In order to arrive at our unified CRC classifier, we pooled all CRC/control data and trained/evaluated using blocked cross-validation as explained above.

Classifiers for adenoma detection

We pursued two strategies for microbiome-based classification of pre-cancerous lesions. First, we evaluated how classifiers trained to distinguish CRC from controls generalized to adenoma (AD) samples. Second, we trained classifiers to differentiate AD from controls.

To assess CRC classifier performance on adenomas, we first performed LOSO validation across all datasets containing AD samples. Classifiers were trained on CRC and control samples from all datasets except the held-out study and then evaluated on adenoma and control samples from that study using the *SIAMCAT* R package (v2.7.2).²⁶ This approach directly tested cross-study generalization and assessed whether CRC-trained models predict AD samples. In addition, we evaluated adenoma detection using the unified CRC classifier by preserving the same training folds but replacing CRC samples in each test fold with adenoma samples, which were randomly and evenly distributed across test folds while control samples remained unchanged. To enable this evaluation, the random forest model was trained outside of the *SIAMCAT* framework using the *ranger* R package⁷³ (v0.16.0) and tested on adenoma and control samples normalized using frozen parameters learned during model training. Performance was quantified using the area under the receiver operating characteristic curve (AUC), treating adenoma as the positive class and controls as the negative class, with AUCs computed using the *pROC* R package⁷⁵ (v1.18.5).

To assess whether adenoma detection benefits from adenoma specific training, we trained classifiers specifically distinguish AD from control samples. These models were evaluated using LOSO validation across all adenoma including studies, alongside a unified adenoma classifier trained on pooled samples and evaluated by 10 fold CV. These analyses enabled a direct comparison between CRC and adenoma specific classifiers for adenoma detection.

Comparison with FIT/FOBT screening

For studies with available data on fecal immunochemical test (FIT) or FOBT results, we compared microbiome-based predictions against these commonly applied screening tests. To this end we calculated the TPR and FPR using FIT/FOBT status as the predictor and CRC status as the outcome for each study.

Feature importance analysis

In order to obtain information about which features contributed most to a model's prediction, we computed Shapley Additive Explanation (SHAP) values. Specifically, for a given training/test split, we used the *kernelshap* function from the *kernelshap* R package⁷⁴ (v0.7.0), supplying the trained model and the profiles of the testing set.

Supplemental information

**Meta-analysis reveals microbiome signatures
for colorectal cancer that are universal
across age groups and sequencing methods**

Selin Pekel, Nicolai Karcher, Morgan Essex, Fabian Springer, Stefano Romano, Quinten R. Ducarmon, Christian Schudoma, Martin Larralde, Alberto Lupatin, Sebastian Zeissig, Michael Zimmermann, and Georg Zeller

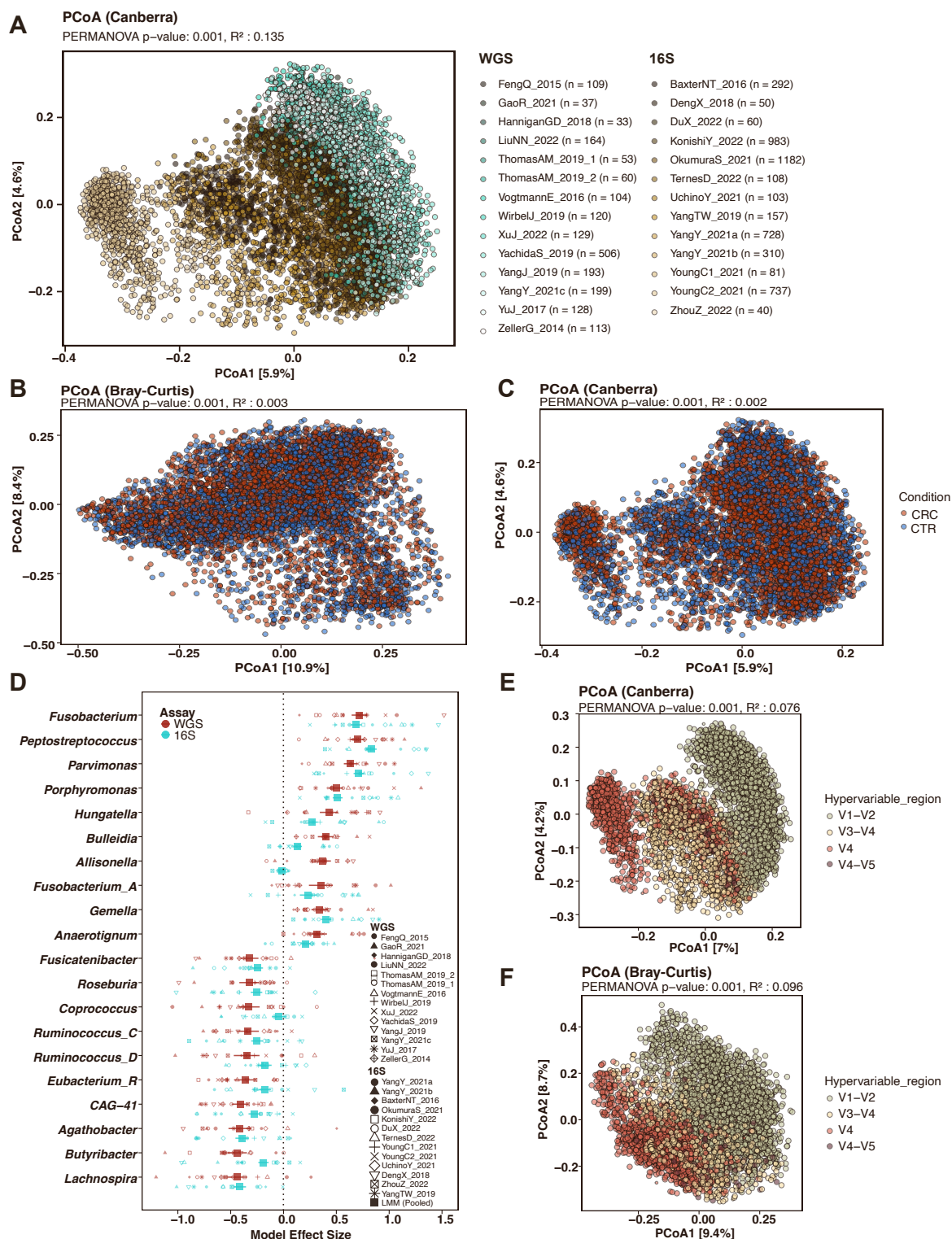
Table S1. Overview of the studies included in the meta-analyses, related to STAR Methods

Dataset	Accession	Country	Assay	Sample Type	Numbers of Samples Included		
					CRC #, +	CTR	AD
BaxterNT_2016 ¹	PRJNA290926	USA	16S	Fecal	120 [#] (100 / 20)	172	198
DengX_2018 ²	PRJNA464414	China	16S	Fecal	17 [#]	33	-
DuX_2022 ³	PRJNA824020	China	16S	Fecal	27 [#] (22 / 5)	33	-
KonishiY_2022 ⁴	DRA012966 DRA013152	Japan	16S	Fecal	546 [#] (471 / 75)	437	240
OkumuraS_2021 ⁵	DRA011735	Japan	16S	Fecal	669 [#] (583 / 86)	513	-
TernesD_2022 ⁶	PRJEB46665	Luxembourg	16S	Fecal	46 [#] (45 / 1)	62	-
UchinoY_2021 ⁷	PRJDB11845	Japan	16S	Fecal	52 [#]	51	-
YangTW_2019 ⁸	PRJNA430990	Taiwan	16S	Fecal	37 [#]	120	126
YangY_2021a ⁹	PRJNA763023	China	16S	Fecal	377 [#] (233 / 144)	351	-
YangY_2021b ⁹	PRJNA763023	China	16S	Fecal	187 [#] (146 / 41)	123	-
YoungC1_2021 ¹⁰	PRJEB36789	India, Vietnam, Chile, Argentina	16S	Fecal	41 [#] (38 / 3)	40	-
YoungC2_2021 ¹¹	PRJEB37635	UK	16S	Fecal	434 [#] (434 / 0)	303	668
ZhouZ_2022 ¹²	PRJNA763872	China	16S	Fecal	22 [#] (18 / 4)	18	22
FengQ_2015 ¹³	PRJEB7774	Austria	WGS	Fecal	46 [#] (42 / 4)	63	47
GaoR_2021 ¹⁴	PRJNA514108	China	WGS	Fecal	16 [#]	21	11
HanniganGD_2018 ¹⁵	PRJNA389927	Canada, US	WGS	Fecal	17 [#] (14 / 3)	16	9
LiuNN_2022 ¹⁶	PRJNA731589	China	WGS	Fecal	79 [#] (63 / 16)	85	-

ThomasAM_2019a ¹⁷	PRJNA447983	Italy	WGS	Fecal	29 [#] (29 / 0)	24	27
ThomasAM_2019b ¹⁷	PRJNA447983	Italy	WGS	Fecal	32 [#] (28 / 4)	28	-
VogtmannE_2016 ¹⁸	PRJEB12449	USA	WGS	Fecal	52 [#] (44 / 8)	52	-
WirbelJ_2019 ¹⁹	PRJEB27928	Germany	WGS	Fecal	60 [#] (53 / 7)	60	-
XuJ_2022 ²⁰	PRJNA758208	China	WGS	Fecal	30 [#]	99	94
YachidaS_2019 ²¹	DRA006684 DRA008156	Japan	WGS	Fecal	257 [#] (225 / 32)	249	67
YangJ_2020 ²²	PRJNA429097	China	WGS	Fecal	98 [#] (67 / 31)	95	-
YangY_2021c ⁹	PRJNA763023	China	WGS	Fecal	99 [#] (49 / 50)	100	-
YuJ_2017 ²³	PRJEB10878	China	WGS	Fecal	74 [#] (70 / 4)	54	-
ZellerG_2014 ²⁴	PRJEB6070	France	WGS	Fecal	52 [#] (49 / 3)	61	42
DrewesJL_2017 ²⁵	PRJNA325649 PRJNA325650	Malaysia	16S	Tissue	86 ⁺ (44 / 42)	-	-
HaleVL_2018 ²⁶	PRJNA445346	USA	16S	Tissue	278 ⁺ (91 / 87)	-	-
NardelliC_2021 ²⁷	PRJNA736583	Italy	16S	Tissue	40 ⁺ (20 / 20)	-	-
PanHW_2020 ²⁸	PRJNA288419	China	16S	Tissue	46 ⁺ (23 / 23)	-	-
TranHNH_2022 ²⁹	PRJNA791834	Vietnam	16S	Tissue	86 ⁺ (43 / 43)	-	-
ZellerG_2014 ²⁴	PRJEB6070	France	16S	Tissue	96 ⁺ (48 / 48)	-	-
ZwinsovaB_2021 ³⁰	PRJEB35990	Czech Republic	16S	Tissue	372 ⁺ (186 / 186)	-	-

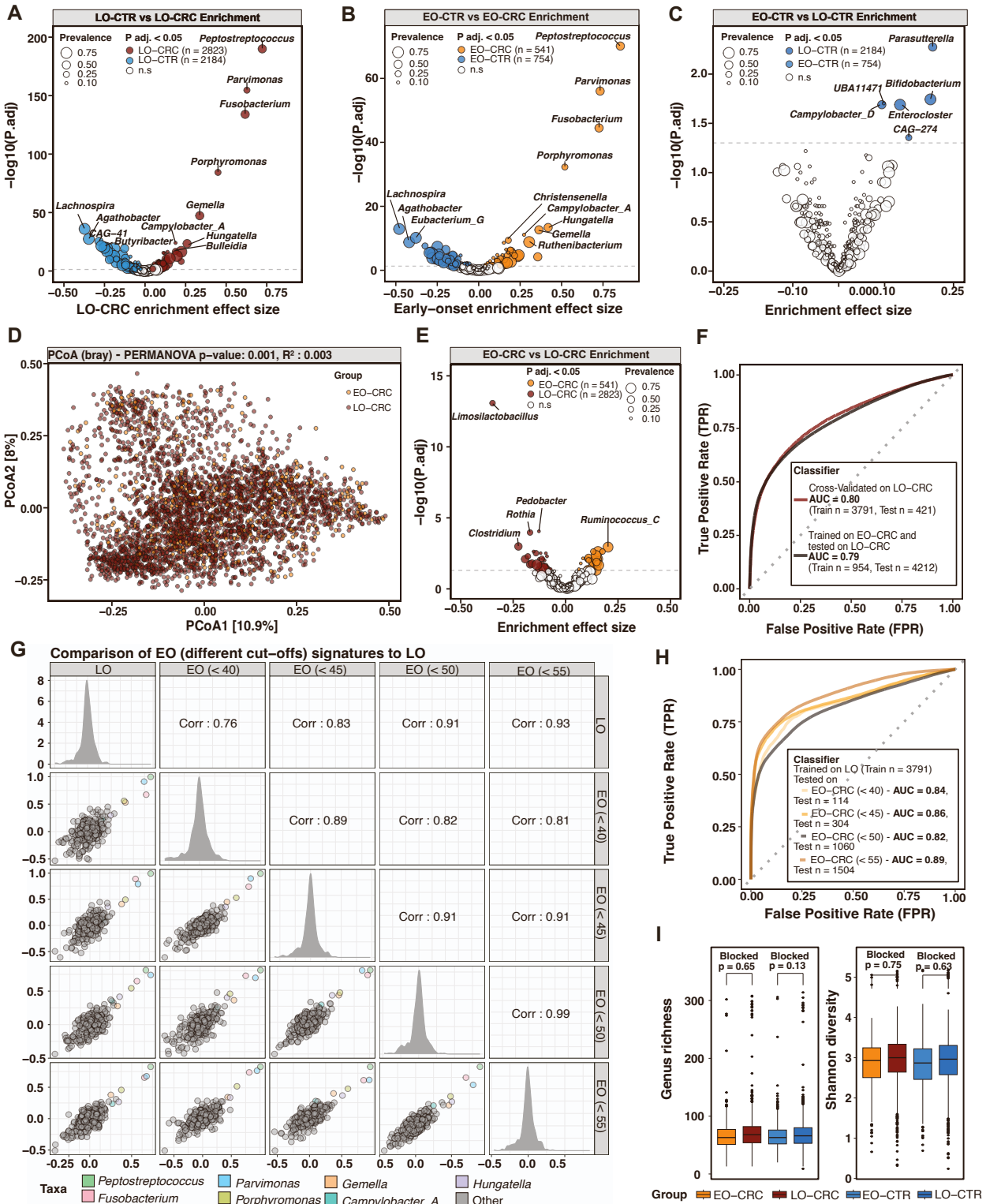
Sequenced stool and tissue samples and available metadata were downloaded from public repositories for

34 study populations (see **STAR Methods**). Samples came from either diagnosed colorectal cancer patients (CRC), patients with adenomas (AD), or control individuals (CTR) and were collected at the time of diagnosis, before treatment. Where age information was available, the breakdown of CRC samples into early-onset (EO) and late-onset (LO) is noted. Letters in the dataset names (a or b) denote independent cohorts within the same study, which were treated as separate datasets in the meta-analysis. Superscripts^{#,+} delineate whether the included samples are split by EO/LO (#) or tumor/adjacent non-tumor (+).

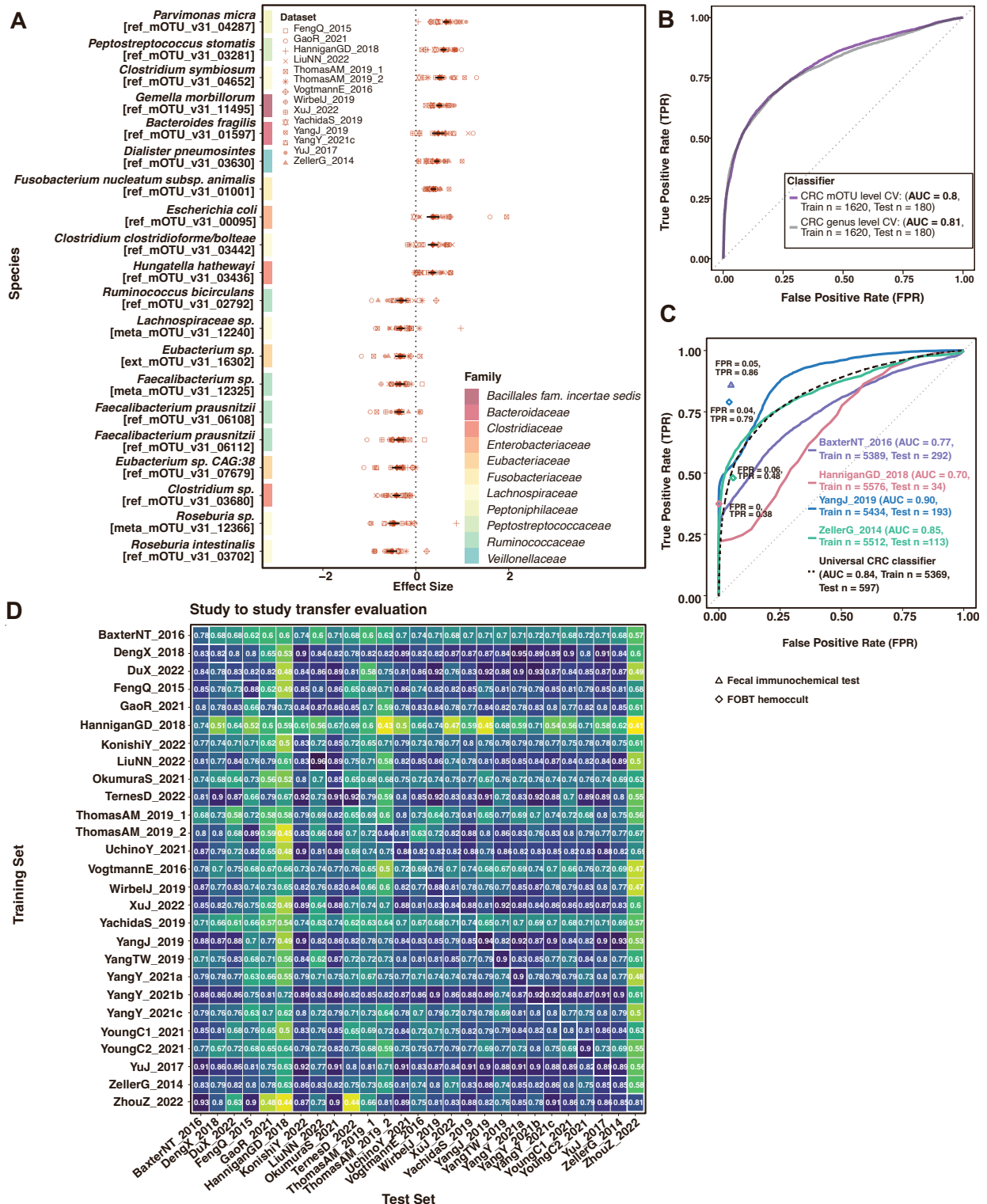


Supplementary Figure 1. Heterogeneity due to study and sequencing approach across datasets included in the meta-analysis. Related to main text **Figure 1**. (A), Principal Coordinates Analysis (PCoA) based on Canberra dissimilarities of all gut microbiome samples used in this study ($n = 6,779$). (B-C), PCoA plots colored by cancer status (CRC vs control samples) using Bray-Curtis (B) and Canberra (C) dissimilarity, respectively. (D), Study-specific and pooled effect sizes of CRC-associated genera (**STAR**

Methods), separated by sequencing approach. (E-F), PCoA of 16S datasets colored by the amplified hypervariable region (V1–V2, V3–V4, V4, V4–V5), based on Canberra (E) and Bray-Curtis (F) distances, reflecting technical variation linked to primer choice.

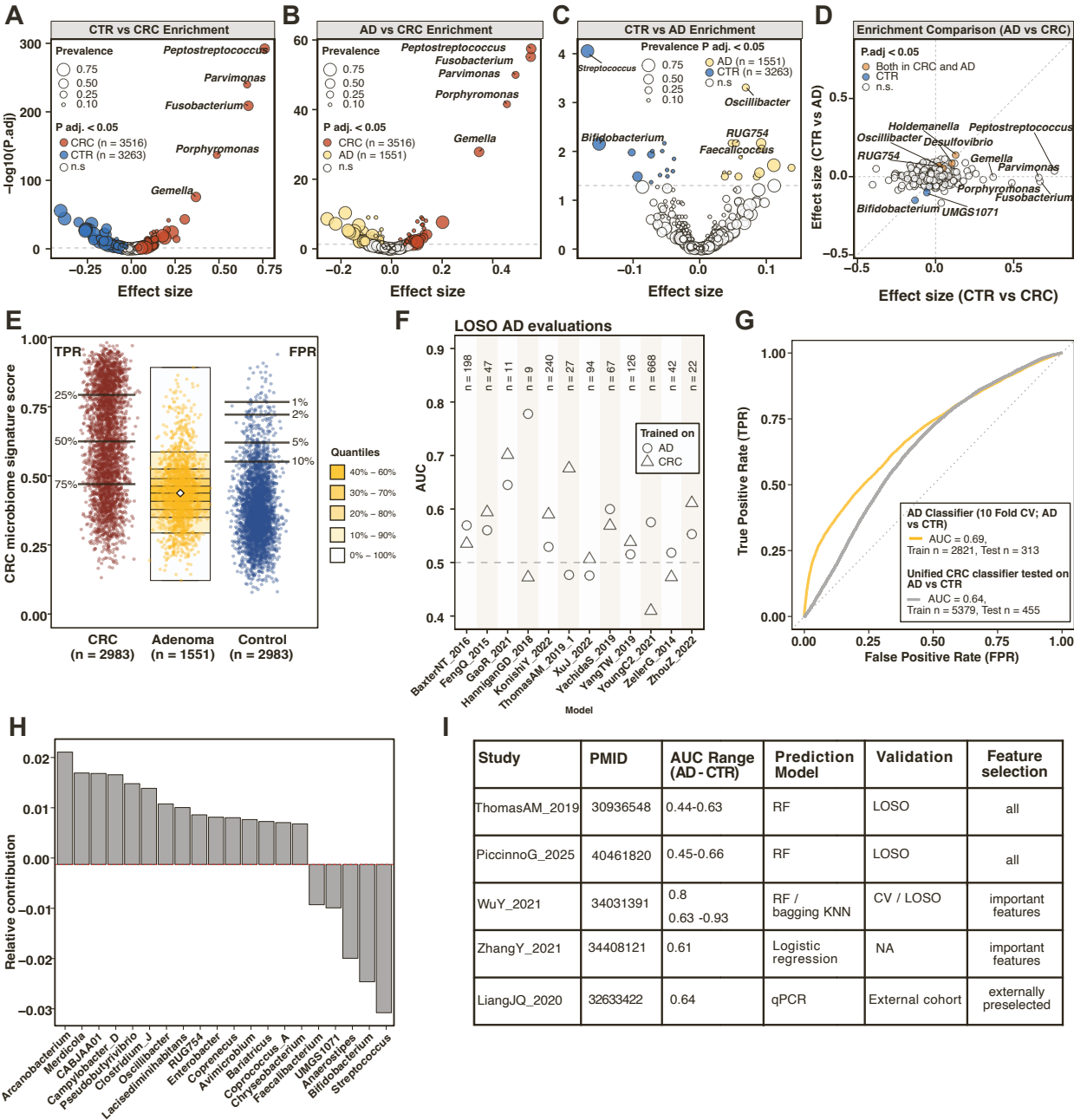


Supplementary Figure 2. Comprehensive meta-analyses of n = 347 genera across early and late onset CRC groups. Related to main text **Figure 2**. Meta-analysis was performed using a linear mixed-effect model (LMM) approach to account for study heterogeneity in the pooled datasets (**STAR Methods**). (A), Differentially abundant genera in late-onset CRC, relative to age-stratified controls (n = 2,823 LO-CRC and n = 2,184 LO-CTR with ≥ 50). (B), Differentially abundant genera in early-onset CRC, relative to age-stratified controls (n = 541 EO-CRC and n = 754 EO-CTR, age < 50). (C), Enrichment in early-onset controls, relative to late-onset controls (n = 756 EO-CTR and n = 2,182 LO-CTR). (D), Principal Coordinates Analysis (PCoA) based on Bray-Curtis dissimilarities comparing early- and late-onset CRC samples. (E), Differentially abundant genera in early-onset CRC relative to late-onset CRC (n = 541 EO-CRC and n = 2,823 LO-CRC). (F), ROC curves of CRC classifiers cross-validated on LO-CRC samples (dark red) or trained on all EO-CRC samples and tested on all LO-CRC samples (black). (G), Pairwise comparisons of early- and late-onset samples using different age cut-offs for early onset (< 40, < 45, < 50, and < 55 years), showing no major age-dependent differences (< 50 used in the main text). CRC-associated taxa are highlighted. (H), Performance of the classifier trained on late-onset CRC and evaluated on early-onset CRC defined by different age cutoffs. (I), Genus richness and Shannon-diversity assessed by comparing EO-CRC with EO-CTR and LO-CRC with LO-CTR using blocked Wilcoxon test (accounting for study) (see **STAR Methods**).



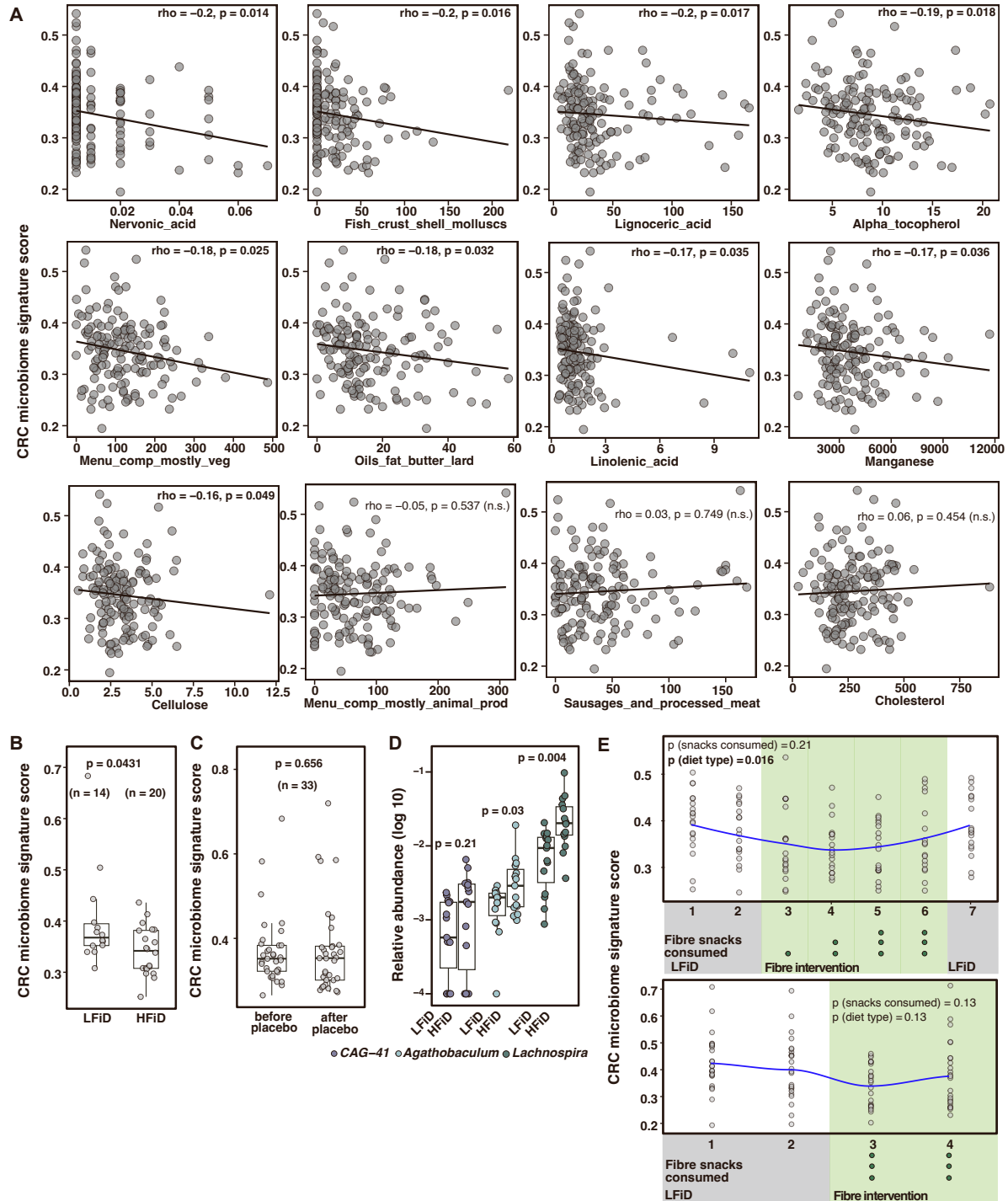
Supplementary Figure 3. Additional validation of the CRC microbiome signature. Related to main text Figure 3. (A), Effect sizes of species (mOTU)-level features associated with CRC across individual datasets. (B), ROC curves comparing cross-validation performance of CRC classifiers trained on mOTU-level vs. genus-level abundance profiles. (C), Performance comparison of metagenomic CRC (genus-level) classifiers with clinically used non-invasive CRC screening tests on the studies where such

data was available. The colored ROC curves indicate LOSO performances with color-matched diamonds and triangles indicating clinical test results (see inset of color key). The dashed black line denotes global performance of the unified CRC classifier for comparison (see main text Fig. 3). (D), Study-to-study classifier transfer performance, whereby each cell shows the area under the receiver operating characteristic curve (AUC) for a model trained on the dataset listed on the y-axis and tested on the dataset listed on the x-axis. Diagonal cells (white outlines) represent within-study performance using cross-validation.



Supplementary Figure 4. Comparison of microbiome signatures and classification performance between adenoma and colorectal cancer. Related to main text **Figures 3 and 4.** (A), Differentially abundant genera between CRC (n = 3,516) and control samples (n = 3,263). Effect sizes are plotted against

$-\log_{10}$ adjusted p values, highlighting significantly CRC-enriched (orange) and CRC-depleted (blue) genera; point size indicates prevalence across datasets. (B), Differentially abundant genera between AD (n = 1,551, yellow) and CRC (n = 3,516, red). (C), Differentially abundant genera between CTR (n = 3,263, blue) and AD (n = 1,551, yellow), with significant taxa colored by enrichment. (D), Scatter plot comparing overall genus-level enrichments obtained between CRC and CTR to those obtained between AD and CTR samples; colored points indicate taxa significantly altered in both CRC and AD (light orange) or in controls (blue). (E), Distribution of CRC microbiome signature scores (median prediction scores from unified model (see **STAR Methods**)) in CRC (n = 2,983), AD (n = 1,551), and control (n = 2,983) samples. Quantile ranges are indicated for AD samples. True positive rates (TPR) at selected thresholds are shown on the lefthand side, corresponding false positive rates (FPR) in controls are shown on the righthand side. (F), LOSO evaluations of AD classification performance across cohorts, showing AUCs for models trained specifically to discriminate between AD and CTR (circles) versus predictions on AD made by the universal CRC classifier (triangles). The number of adenoma samples per study is indicated on top. (G), AUROC curves for a 10-fold cross-validation classifier trained to discriminate between AD and CTR as well as for the unified CRC classifier tested on AD vs CTR samples. For the latter, test folds were identical to those used for the unified CRC model but all CRC samples were replaced by adenoma samples. (H), Relative contribution of individual genera to the unified adenoma classifier shown in panel G (orange), quantified using mean absolute SHAP values and scaled to represent each feature's relative contribution to the model. Only the top 20 features (by relative contributions to the model) are shown. (I), Summary of previously published AD microbiome classifiers, including study, PMID, reported AUC ranges, prediction models used, validation strategies, and feature selection approaches^{31–35}.



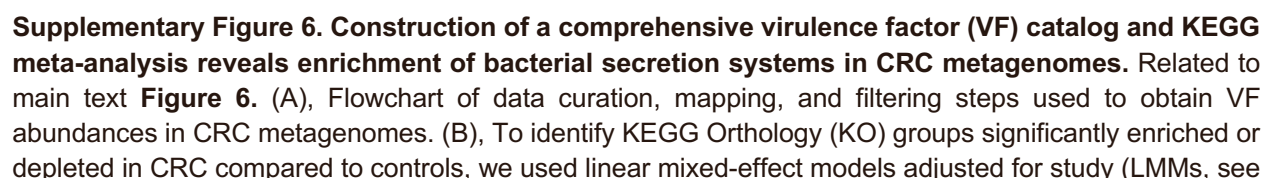
Supplementary Figure 5. Broader dietary associations and dietary intervention-related changes in CRC microbiome signature scores. Related to main text **Figure 5**. (A), Correlations between CRC microbiome signature scores and dietary metadata in healthy, tumor-free overweight individuals at baseline in the Sowah et al. ³⁶. (B), Baseline dietary fiber intake comparison of samples reported in the Healey et al. ³⁷, stratified into low-fiber diet (LFID) and high-fiber diet (HFID) groups. (C), CRC microbiome signature scores before and after placebo intervention in the Healey study ³⁷. (D), The dietary intervention underlying the reduction in CRC microbiome signature scores shown in Fig. 5D was accompanied by pronounced

A

```
graph LR
    PubMed[Literature PubMed n = 51] --> BLAST[BLAST n = 188]
    GenBank[GenBank] --> BLAST
    BLAST --> Curate[Curate]
    Curate --> VFDB[VFDB 2.0 n = 34429]
    Curate --> Ext[Ext. VFs n = 239]
    VFDB --> Map[Map DIAMOND]
    Ext --> Map
    Map --> GMGC[GMGC genes mapped to VFs n = 49438]
    GMGC --> Filter[Filter]
    Filter --> KEGG[KEGG-filtered GMGC genes mapped to VFs n = 11706]
```

Flowchart illustrating the bioinformatics pipeline for identifying novel viral proteins:

- Literature PubMed** (n = 51) and **Homologs** (n = 188) are compared using **BLAST**.
- The results are **Curate**d into **VFDB 2.0** (n = 34429) and **Ext. VFs** (n = 239).
- The sequences are **Map**ped using **DIAMOND** to **GMGC genes mapped to VFs** (n = 49438).
- The results are **Filter**ed to yield **KEGG-filtered GMGC genes mapped to VFs** (n = 11706).



STAR Methods) with an FDR cutoff < 0.05. For visualization, significant KOs were uploaded to KEGG Mapper for the pathway ko03070: bacterial secretion system. KOs were colored by enrichment in CRC (red), depletion in CRC (blue), or non-significant associations (grey). KOs shown in white were not tested (not detected in the data).

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